DATASTRUCTURES AND ALGORITHMS

23CSE203

LAB MANUAL

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Week-1

A.AIM: A rocket launch system displays a countdown before launch. Instead of using a loop implement the countdown using recursion. Write a recursion python program to display a countdown from n to 1 followed by “launch”.

PROGRAM:

```
def launch(n):
    for i in range (n, -1,-1):
        if i==0:
            print("launch")
        else:
            print(i)
n=int (input ("Enter no. of seconds to launch:"))
launch(n)
```

OUTPUT:



EXPLANATION:

This program displays a countdown from n to 1 and finally prints 'launch'. Although the aim mentions recursion, the given code uses a loop.

ERROR TABLE:

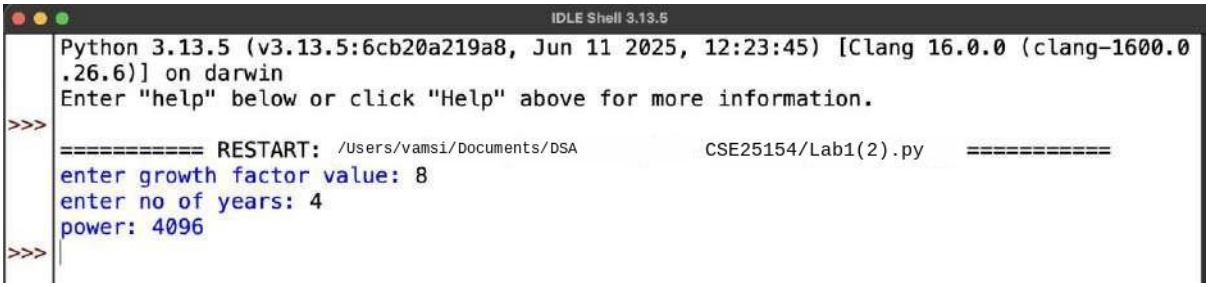
S.no.	Errors	Rectification
1.	Indentation Error	Use 4 spaces for each level of indentation.
2.	Type Error	Convert the input to an integer using n = int (input ("Enter no. of seconds to launch: ")).

B.AIM: The banks calculate the growth of an investment using compound interest; the formula requires calculating powers. Which can be implemented recursively. Write a recursive program to calculate p^n(power), where p=principal growth factor, n= number of years.

PROGRAM:

```
p = float (input ("Enter principal growth factor (p): "))
n = int (input ("Enter number of years (n): "))
def power (p, n):
    if n == 0:
        return 1
    else:
        return p * power (p, n - 1)
result = power (p, n)
print ("Power: ", result)
```

OUTPUT:



EXPLANATION:

- The function of power `(p, n)` calculates p^n using recursion by multiplying `p` with the result of power `(p, n-1)`.
- The recursion stops when `n` becomes `0`, returning `1`, which is the base case.
- The final result is displayed after all recursive calls are completed.

ERROR TABLE:

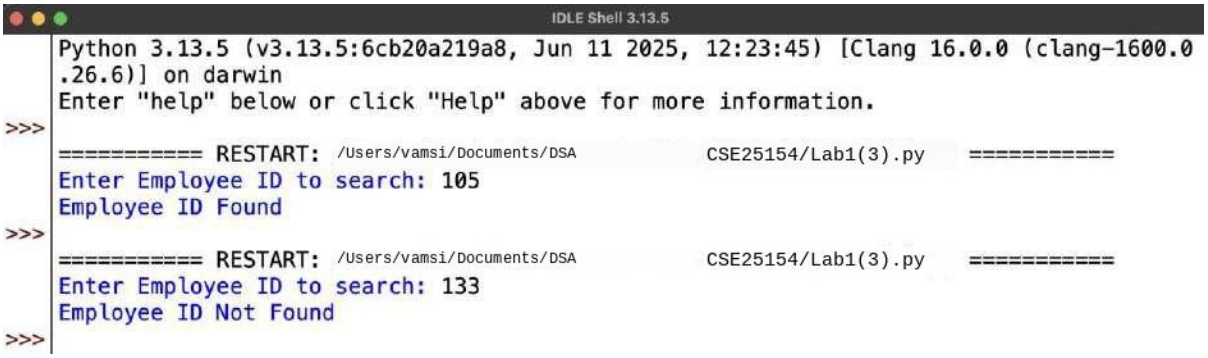
S.no.	Errors	Rectification
1.	Type Error	Convert the input to float or int using float (input()) or int (input ()).
2.	Incorrect output if the recursive call is written as power (p, n + 1) instead of power (p, n - 1).	Change the recursive call to power (p, n - 1) so that n decreases toward the base case.

C.AIM: An HIR department stores employee IDs in a list. Develop a recursive program to search for a given employee ID. Write a recursive function to search for an employee ID in a list.

PROGRAM:

```
def search (list, key, index):
    if index >= len(list):
        return False
    if list[index] == key:
        return True
    return search (lst, key, index + 1)
employees = [101, 102, 103, 104, 105]
key = int (input ("Enter Employee ID to search: "))
if search (employees, key, 0):
    print ("Employee ID Found")
else:
    print ("Employee ID Not Found")
```

OUTPUT:



EXPLANATION:

- The recursive function search () checks each employee ID in the list one by one, starting from the given index.
- If the current employee ID matches the required ID (key), it returns True.
- If the ID does not match, the function calls itself recursively with the next index (index + 1).

- If the index reaches the end of the list, it returns False, indicating that the employee ID was not found.
- Finally, the if-else statement displays whether the employee ID is **Found** or **Not Found**.

ERROR TABLE:

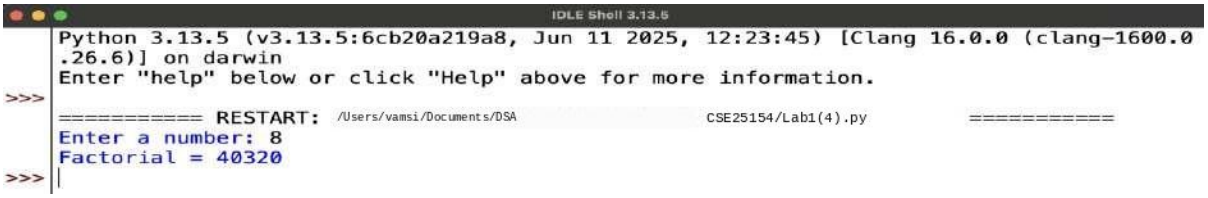
S.no.	Errors	Rectification
1.	Name Error if the list name is misspelled (e.g., lst instead of list).	Use the correct variable name consistently throughout the program.
2.	Syntax Error due to a missing colon (:) after an if statement or function definition.	Add the missing colon (:) after if, else, or def.

D.AIM: In a logistics company the number of possible ways to arrange n different parcels are calculated using the factorial formula(n!). develop a recursive program to calculate the factorial of a given number.

PROGRAM:

```
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial (n - 1)
num = int (input ("Enter a number: "))
if num < 0:
    print ("Factorial does not exist for negative numbers.")
else:
    print ("Factorial =", factorial(num))
```

OUTPUT:



EXPLANATION:

- The function `factorial(n)` computes the factorial by multiplying the current number with the factorial of the previous number.
- The recursion ends when `n` becomes `0` or `1`, where the factorial value is `1`.
- The program also checks for negative inputs since factorials are defined only for non-negative integers.

ERROR TABLE:

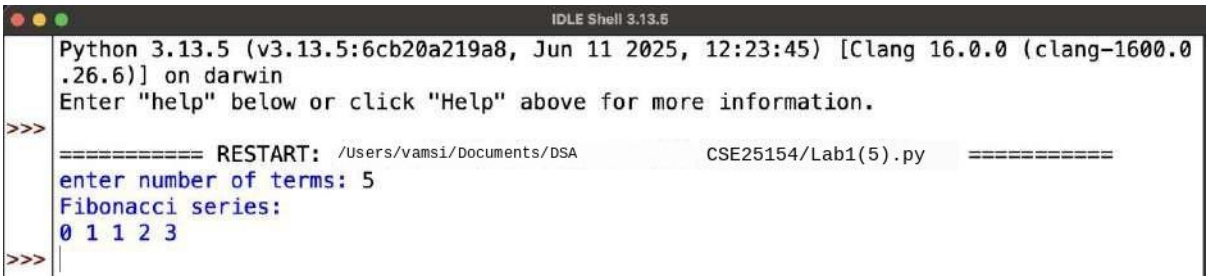
S.no	Errors	Rectification
1.	**Name Error** occurs when a variable or function is used without defining it first.	Define the variable or function before using it in the program.
2.	**Value Error** occurs when an invalid value, such as a letter, is entered where a number is expected.	Enter a valid numerical value or use proper input validation.

E.AIM: Write a recursive python program to print the first n terms of the Fibonacci series.

PROGRAM:

```
n = int (input ("enter number of terms: "))
def fibonacci(num):
    if num==0:
        return 0
    elif num==1:
        return 1
    else:
        return fibonacci(num-1) +fibonacci(num-2)
print ("Fibonacci series: ")
for i in range(n):
    print(fibonacci(i), end=" ")
```

OUTPUT:



EXPLANATION:

- The function Fibonacci(n) returns the nth Fibonacci number by recursively adding the previous two Fibonacci numbers.
- The base cases return 0 for the first term and 1 for the second term.
- A for loop calls the recursive function repeatedly to print the first n terms of the Fibonacci series. terms of the Fibonacci series.

ERROR TABLE:

S.no.	Errors	Rectification
1.	Recursion Error if the base cases (n == 0 and n == 1) are not included.	Add both base conditions before the recursive call.
2.	Value Error if the user enters text instead of an integer.	Enter a valid integer for the number of terms.

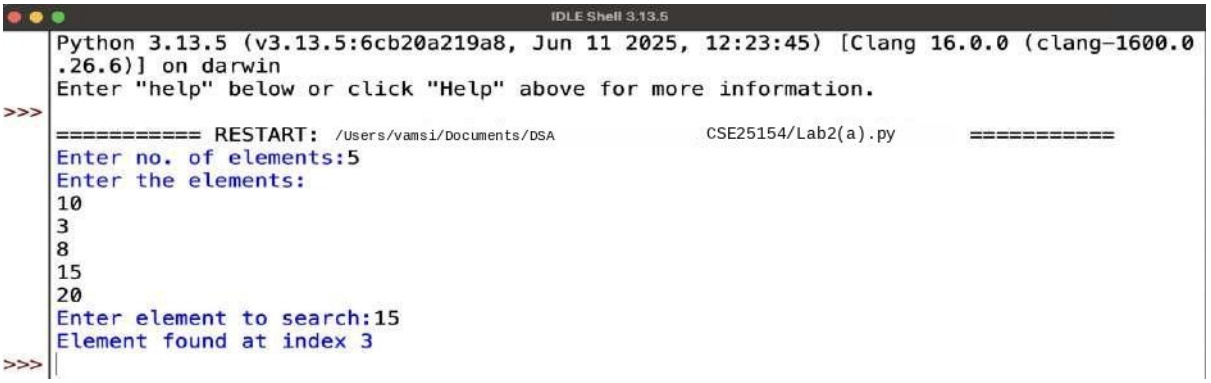
Week-2

A.AIM: Write a program to perform linear search on a given input.

PROGRAM:

```
def linear_search(arr,key):
    for i in range(len(arr)):
        if arr[i]==key:
            return i
    return -1
 n = int (input ("Enter no. of elements:"))
arr= []
print ("Enter the elements:")
for i in range(n):
    arr.append(int(input ()))
key=int (input ("Enter element to search:"))
result=linear_search(arr,key)
if result! =-1:
    print ("Element found at index", result)
else:
    print ("Element not found.")
```

OUTPUT:



EXPLANATION:

- The linear_search() function checks each element of the list one by one and compares it with the given `key`.
- If the key is found, its index is returned; otherwise, the function returns `-1`.
- The program takes the array and search element as input and displays whether the element is found.

ERROR TABLE:

S.no.	Errors	Rectification
1.	ValueError occurs if a non-integer value is entered when int (input ()) is used.	Enter only integer values for the number of elements and array elements.
2.	TypeError may occur if len(arr) is used on a variable that is not a list.	Make sure arr is initialized as a list using arr = [] and elements are appended to it.

B(i). AIM: Write a python program to implement binary search with sorted array as an input.

PROGRAM:

```
def binary_search(arr, key):
    low = 0
    high = len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return -1
n = int (input ("Enter number of elements: "))
arr = []
print ("Enter elements in sorted order:")
for i in range(n):
    arr.append(int(input ()))
key = int (input ("Enter the element to search: "))
result = binary_search(arr, key)
if result != -1:
    print("Element found at index", result)
else:
```

```
print("Element not found")
```

OUTPUT:

```
========== RESTART: /Users/vamsi/Documents/DSA          CSE25154/Lab2(b(i)).py    =========
Enter number of elements: 5
Enter elements in sorted order:
11
22
33
44
55
Enter the element to search: 44
Element found at index 3
>>>
========== RESTART: /Users/vamsi/Documents/DSA          CSE25154/Lab2(b(i)).py   =========
Enter number of elements: 5
Enter elements in sorted order:
11
22
33
44
55
Enter the element to search: 66
Element not found
>>>
```

EXPLANATION:

- The program defines a binary_search() function that takes a sorted list and a search element as input.
- The variables low and high are initialized to represent the first and last positions of the list.
- The program calculates the middle position using mid = (low + high) // 2 and compares the middle element with the search element.
- If the search element is greater or smaller than the middle element, the program searches only the required half of the list.
- If the element is found, its index is returned; otherwise, -1 is returned and **"Element not found"** is displayed.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**Indentation Error** if the statements inside the function or loop are not properly indented.	Maintain proper indentation inside the function and loop.
2.	**ValueError** if a non-numeric value is entered instead of an integer.	Enter only valid integer values. valid integer values.
3.	Incorrect result if the elements are not entered in sorted order.	Enter the elements in sorted order before performing binary search.
4.	**IndexError** if an invalid index is accessed while searching.	Make sure low and high remain within the list range.

B(ii). AIM: Write a python program to implement binary search without sorted(unsorted) array as an input

PROGRAM:

```
def binary_search(arr, key):
    arr.sort()
    low = 0
    high = len(arr) - 1
     while low <= high:
        mid = (low + high) // 2
         if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid – 1
    return -1
 n = int (input ("Enter number of elements: "))
arr = []
print ("Enter elements:")
for i in range(n):
     arr.append(int(input()))
key = int (input ("Enter the element to search: "))
result = binary_search(arr, key)
 if result != -1:
    print ("Element found at index", result)
```

```
else:
    print ("Element not found")
```

OUTPUT:



EXPLANATION:

1. The binary_search() function is defined to search for the given key in the list arr.
2. The arr.sort() function arranges the elements in ascending order so that binary search can be performed.
3. The variables low and high represent the first and last positions of the sorted list.
4. The program finds the middle element and compares it with the key. Based on the comparison, it searches either the left half or the right half.
5. If the element is found, its index is returned. If the element is not found, the function returns -1.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**IndentationError** may occur if while, if, or other statements are not properly aligned.	Maintain proper indentation inside the function and conditional blocks.
2.	**SyntaxError** may occur if – is used instead of the normal minus operator - in high = mid – 1.	Use the normal hyphen: high = mid - 1.
3.	**ValueError** occurs if a non-numeric value is entered instead of an integer.	Enter only valid integer values. valid integer values.
4.	**NameError** occurs if variables such as arr, key, or result are misspelled.	Check the spelling and define variables before using them.

Week-3

A.AIM: Write a python program to implement bubble sort, where the input elements are taken from the user.

PROGRAM:

```
#bubble sort
def bubble_sort(arr):
    n = len(arr)
    for i in range (n-1):
        for j in range(n-i-1):
            if arr[j]>arr[j+1]:
                arr[j],arr[j+1] = arr[j+1],arr[j]
    return arr
#taking input
n = int(input("Enter number of elements: "))
arr=[]
print("Enter elements: ")
for i in range(n):
    arr.append(int(input()))
bubble_sort(arr)
print("sorted array: ")
for element in arr:
    print(element, end=" ")
```

OUTPUT:

```
>>>
===== RESTART: /Users/vamsi/Documents/DSA CSE25154/Lab3(1) bubble sort.py =====
Enter number of elements: 5
Enter elements:
12
65
33
25
1
sorted array:
1 12 25 33 65
>>>
```

EXPLANATION:

- The bubble_sort() function is defined to arrange the elements of the list in ascending order.
- The outer for loop controls the number of passes, while the inner for loop compares adjacent elements.
- The condition arr[j] > arr[j+1] checks whether the current element is greater than the next element.
- If the elements are in the wrong order, they are exchanged using arr[j], arr[j+1] = arr[j+1], arr[j].
- After all comparisons and swaps are completed, the sorted array is returned and displayed element by element.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**ValueError** if a non-integer value is entered as an array element.	Enter only valid integer values and linely
2.	**NameError** if arr, n, or another variable is misspelled.	Check the variable names and use the correct spelling.
3.	**TypeError** may occur if the list contains incompatible data types during comparison.	Enter elements of the same suitable data type, such as integers.
4.	**IndexError** may occur if the program accesses an invalid array position.	Use the correct loop range range(n-i-1) to keep the index within the list.

B.AIM: Write a python program to implement insertion sort, where the input elements are taken from the user.

PROGRAM:

```
def insertion_sort(arr): n = len(arr) for i in range (1, n): key = arr[i] j = i - 1

   while j >= 0 and arr[j] > key:
       arr[j + 1] = arr[j]
       j = j - 1
    arr[j + 1] = key
return arr
#Taking input

n = int (input ("Enter number of elements: ")) arr = []

print ("Enter elements: ")

for i in range(n):

     arr.append(int(input()))

#Sorting the array

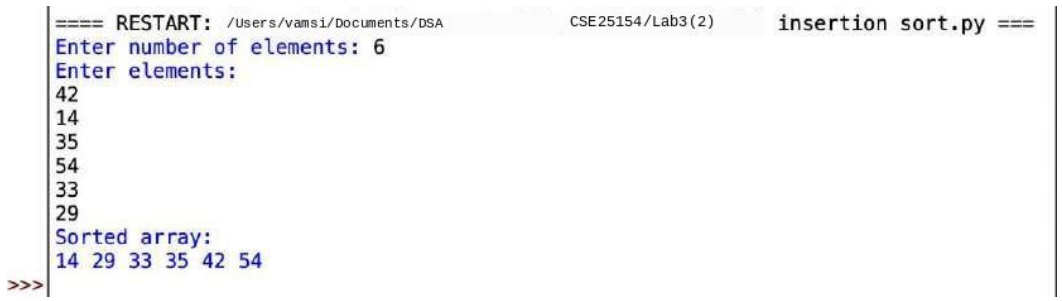
insertion_sort(arr)

print ("Sorted array: ")

for element in arr:

       print (element, end=" ")
```

OUTPUT:



EXPLANATION:

- The insertion_sort() function is defined to arrange the elements of the list in **ascending order**.
- The for loop starts from the second element because the first element is considered already sorted.
- The variable key stores the current element, and j is used to compare it with the previous elements.
- The while loop shifts elements that are greater than key one position to the right.
- The key is placed in its correct position, and after all elements are processed, the sorted array is displayed.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**IndentationError** if the statements inside the function, for, or while loop are not properly indented.	Maintain proper indentation inside all blocks.
2.	**ValueError** if a non-integer value is entered as an array element.	Enter only valid integer values.
3.	**NameError** if variables such as arr, key, or j are misspelled.	Check the variable names and use the correct spelling.
4.	**IndexError** if an invalid list index is accessed while shifting elements.	Make sure j >= 0 is checked before accessing arr[j].

C.AIM: Write a python program to implement selection sort, where the input elements are taken from the user.

PROGRAM:

```
#Selection Sort

def selection_sort(arr): n = len(arr)

for i in range(n - 1):
    min_index = i
    for j in range(i + 1, n):
        if arr[j] < arr[min_index]:
            min_index = j
    arr[i], arr[min_index] = arr[min_index], arr[i]
return arr


#Taking input

n = int(input("Enter number of elements: "))

arr = [] print("Enter elements: ")

for i in range(n):

    arr.append(int(input()))

#Sorting the array

selection_sort(arr)

print("Sorted array: ")

for element in arr:

        print(element, end=" ")
```

OUTPUT:

```
>>>
==== RESTART: /Users/vamsi/Documents/DSA          CSE25154/Lab3(3)          selection sort.py ===
Enter number of elements: 6
Enter elements:
22
42
12
2
72
32
Sorted array:
2 12 22 32 42 72
>>>
```

EXPLANATION:

- The selection_sort() function is defined to arrange the elements of the list in **ascending order**.
- The outer for loop selects the position where the smallest element should be placed.
- The variable min_index stores the index of the smallest element found in the unsorted part of the list.
- The inner for loop compares the elements and updates min_index whenever a smaller element is found.
- The smallest element is swapped with the element at position i, and after all passes, the sorted array is displayed.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**IndentationError** if the function or loop statements are not properly indented.	Maintain proper indentation inside the function and loops.
2.	**ValueError** if a non-integer value is entered as an array element.	Enter only valid integer values.
3.	**NameError** if variables such as arr, i, j, or min_index are misspelled.	Check the variable names and use the correct spelling.
4.	**IndexError** if an invalid list index is accessed.	Use the correct loop ranges and ensure the index is within the list range.

Week-4

A.AIM: Write a python program to implement merge sort, where the input elements are taken from the user.

PROGRAM:

```
def merge_sort(arr):
    if len(arr) <= 1:
        return arr
    mid = len(arr) // 2
    left = arr[:mid]
    right = arr[mid:]
    left = merge_sort(left)
    right = merge_sort(right)
    i = j = k = 0
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            arr[k] = left[i]
            i += 1
        else:
            arr[k] = right[j]
            j += 1
        k += 1
    while i < len(left):
        arr[k] = left[i]
        i += 1
        k += 1
    while j < len(right):
        arr[k] = right[j]
        j += 1
```

```
        k += 1
    return arr
n = int(input("Enter number of elements: "))
arr = []
print("Enter elements:")
for i in range(n):
    arr.append(int(input()))
merge_sort(arr)
print("Sorted array:")
for element in arr:
    print(element, end=" ")
```

OUTPUT:

```
====== RESTART: /Users/vamsi/Documents/DSA CSE25154/Lab4(a) merge sort.py =====
Enter number of elements: 5
Enter elements:
16
52
22
7
4
Sorted array:
4 7 16 22 52
>>>
```

EXPLANATION:

- The merge_sort() function is defined to arrange the elements of the list in **ascending order** using the Merge Sort algorithm.
- The if len(arr) <= 1 condition checks whether the list has **one or no elements**. If so, the list is already sorted and returned.
- The variable mid stores the **middle index** of the list, which is used to divide the list into two halves.
- The list is divided into two parts: the left contains the first half and the right contains the second half.
- The merge_sort() function is called recursively to **sort both the left and right halves**.
- The variables i, j, and k are used as indexes for the **left list, right list, and original list** respectively.
- The while loop compares the elements of the left and right lists and places the **smaller element** into the original list.
- The next while loops copy any **remaining elements** from the left or right list into the original list.

- The return arr statement returns the **sorted list** after merging the two halves.
- The user enters the **number of elements** and the elements of the list using input().
- The merge_sort(arr) function is called to **sort the given list**.
- Finally, the sorted elements are displayed using a for loop.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**IndentationError** occurred because some statements inside the function and loops were not properly aligned.	Corrected the indentation of the statements according to Python syntax.
2.	**ValueError** occurred when I entered a non-numeric value while giving the array elements.	Entered only integer values as input for the array.
3.	**TypeError** occurred because incompatible data types were present while comparing the elements.	Used integer values of the same suitable data type for sorting.
4.	**Recursion Error** occurred because the recursive function did not reach the stopping condition properly.	Added and verified the base condition if len(arr) <= 1.
5.	**IndexError** occurred when the program tried to access an element outside the valid range of the list.	Corrected the loop conditions and ensured that the indexes remain within the list range.

B.AIM: Write a python program to implement quick sort, where the input elements are taken from the user.

PROGRAM:

```
def quicksort(a, low, high):
    if low < high:
        i = low
        j = high
        pivot = low
        while i < j:
            while i <= high and a[i] <= a[pivot]:
                i += 1
            while a[j] > a[pivot]:
                j -= 1
            if i < j:
                a[i], a[j] = a[j], a[i]
        a[pivot], a[j] = a[j], a[pivot]
        quicksort(a, low, j - 1)
        quicksort(a, j + 1, high)
n = int(input("Enter number of elements: "))
a = []
print("Enter elements:")
for i in range(n):
    a.append(int(input()))
quicksort(a, 0, n - 1)
print("Sorted array:")
for element in a:
    print(element, end=" ")
```

OUTPUT:

```
====== RESTART: /Users/vamsi/Documents/DSA CSE25154/Lab4(b) quicksort.py ======
Enter number of elements: 5
Enter elements:
25
12
81
18
11
Sorted array:
11 12 18 25 81
```

EXPLANATION:

- The quicksort() function is used to arrange the elements of the array in **ascending order**.
- The condition if low < high checks whether the array portion contains more than one element.
- The variables i and j are used to **scan the array from both sides**.
- The variable pivot stores the index of the **first element selected as the pivot**.
- The first while loop moves i forward until an element greater than the pivot is found.
- The second while loop moves j backward until an element smaller than or equal to the pivot is found.
- If i < j, the elements at i and j are **swapped**.
- The pivot is then swapped with the element at j, placing it in its **correct position**.
- The function is recursively called for the **left and right portions** of the pivot.
- The user enters the number of elements and their values, and quicksort() is called to sort the array.
- Finally, the **sorted array is displayed**.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**ValueError** occurred when I entered a non-integer value for an array element.	Entered only valid integer values while providing the input.
2.	**NameError** occurred because I used an incorrect variable name such as pivot, i, or j.	Checked the variable names and corrected their spelling throughout the program.
3.	**Index Error** occurred when the program tried to access an array position outside its valid range.	Corrected the loop conditions and ensured that the indexes remain within the array boundaries.
4.	**Recursion Error** occurred when the recursive calls continued without reaching the proper stopping condition.	Ensured that the condition if low < high correctly controls the recursive calls.

Week-5

A.AIM: Write a python program to implement single linked list with below operations

1. Create Linked list
2. Insert beginning
3. Insert end
4. Insert at index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display/Traverse
10. Exit

PROGRAM:

class Node: def **init**(self, data): self.data = data self.next = None

class SinglyLinkedList: def **init**(self): self.head = None

```
# 1. Create linked list
def create(self):
    n = int(input("Enter number of nodes: "))

    for i in range(n):

        data = int(input(f"Enter data for node {i + 1}: "))
        new_node = Node(data)

        if self.head is None:
            self.head = new_node
        else:
            temp = self.head
            while temp.next:
                temp = temp.next
            temp.next = new_node

    print("Linked list created successfully.")

# 2. Insert at beginning
```

```
def insert_beginning(self):
    data = int(input("Enter data: "))
    new_node = Node(data)

    new_node.next = self.head
    self.head = new_node

    print("Node inserted at beginning.")

# 3. Insert at end
def insert_end(self):
    data = int(input("Enter data: "))
    new_node = Node(data)

    if self.head is None:
        self.head = new_node
    else:
        temp = self.head
            while temp.next:

                temp = temp.next

            temp.next = new_node

         print("Node inserted at end.")

     # 4. Insert at specific index

    def insert_index(self):

        index = int(input("Enter index: "))

        data = int(input("Enter data: "))

        new_node = Node(data)

         if index == 0:

            new_node.next = self.head

            self.head = new_node

            print("Node inserted.")

            return
```

```
        temp = self.head
        for i in range(index - 1):
            if temp is None:
                print("Invalid index.")
                return
            temp = temp.next
        if temp is None:
            print("Invalid index.")
            return
        new_node.next = temp.next
        temp.next = new_node
        print("Node inserted.")
    # 5. Delete by value
    def delete_value(self):
        value = int(input("Enter value to delete: "))
        if self.head is None:
            print("List is empty.")
            return
        if self.head.data == value:
            self.head = self.head.next
            print("Node deleted.")
            return
        temp = self.head

        while temp.next and temp.next.data != value:
```

```
            temp = temp.next
        if temp.next is None:
            print("Value not found.")
        else:
            temp.next = temp.next.next
            print("Node deleted.")
    # 6. Delete first node
    def delete_first(self):
        if self.head is None:
            print("List is empty.")
        else:
            self.head = self.head.next
            print("First node deleted.")
    # 7. Delete last node
    def delete_last(self):
        if self.head is None:
            print("List is empty.")
            return
        if self.head.next is None:
            self.head = None
            print("Last node deleted.")
            return

        temp = self.head
        while temp.next.next:
```

```
            temp = temp.next
        temp.next = None
        print("Last node deleted.")
    # 8. Count nodes
    def count(self):
        count = 0
        temp = self.head
        while temp:
            count += 1
            temp = temp.next
        print("Number of nodes:", count)
    # 9. Display / Traverse
    def display(self):
        if self.head is None:
            print("List is empty.")
            return
        temp = self.head
        while temp:
            print(temp.data, end=" -> ")
            temp = temp.next
        print("None")
# Main program
sll = SinglyLinkedList()
while True:
    print("\n--- SINGLY LINKED LIST ---")
```

```
print("1. Create linked list")
print("2. Insert at beginning")
print("3. Insert at end")
print("4. Insert at specific index")
print("5. Delete by value")
print("6. Delete first node")
print("7. Delete last node")
print("8. Count number of nodes")
print("9. Display / Traverse")
print("10. Exit")
 choice = int(input("Enter your choice: "))
 if choice == 1:
    sll.create()
 elif choice == 2:
    sll.insert_beginning()
  elif choice == 3:
     sll.insert_end()
  elif choice == 4:
    sll.insert_index()
 elif choice == 5:
    sll.delete_value()
 elif choice == 6:
    sll.delete_first()
 elif choice == 7:
    sll.delete_last()
```

```
    elif choice == 8:
        sll.count()
    elif choice == 9:
        sll.display()
    elif choice == 10:
        print("Program exited.")
        break
    else:
        print("Invalid choice.")
```

OUTPUT:



```
Enter your choice: 2
Enter data: 5
Node inserted at beginning.

--- SINGLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 3
Enter data: 55
Node inserted at end.

--- SINGLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 4
Enter index: 3
Enter data: 33
Node inserted.

--- SINGLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 5
Enter value to delete: 100
Value not found.
```

```
--- SINGLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 5
Enter value to delete: 5
Node deleted.

--- SINGLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 6
First node deleted.

--- SINGLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 7
Last node deleted.

--- SINGLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
```

```
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 8
Number of nodes: 5

--- SINGLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 9
18 -> 33 -> 81 -> 2 -> 25 -> None

--- SINGLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 10
Program exited.
```

EXPLANATION:

- The Node class is used to create a **node containing data and a link to the next node**.
- The SinglyLinkedList class maintains the linked list using the head pointer.

- The create() function creates a linked list by adding the given number of nodes.
- The insert_beginning() function adds a **new node at the beginning** of the list.
- The insert_end() function adds a **new node at the end** of the list.
- The insert_index() function inserts a node at a **specified index**.
- The delete_value() function searches for a given value and **deletes the corresponding node**.
- The delete_first() function removes the **first node** from the list.
- The delete_last() function removes the **last node** from the list.
- The count() function counts and displays the **total number of nodes**.
- The display() function traverses the list and **prints all the elements**.
- The main program displays a **menu** and performs the selected linked-list operation.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**Node Error** occurred because the Node class was not defined properly.	Defined the Node class before using it.
2.	**Attribute Error** occurred when temp.next was not handled correctly.	Updated temp using temp = temp.next.
3.	**Index Error** occurred when an invalid index was entered.	Checked whether the index is within the list range.
4.	**Value not found** occurred when the entered value was not present in the list.	Searched the list and displayed **"Value not found."**
5.	**Empty list error** occurred when an operation was performed on an empty list.	Checked self.head is None before performing the operation.

B.AIM: Write a Python program to implement a Double linked list with the above operations.

PROGRAM:

class Node: def **init**(self, data): self.data = data self.prev = None self.next = None

class DoublyLinkedList: def **init**(self): self.head = None

```
# 1. Create linked list
def create(self):
    n = int(input("Enter number of nodes: "))

    for i in range(n):

        data = int(input(f"Enter data for node {i + 1}: "))
        new_node = Node(data)
        if self.head is None:
            self.head = new_node
        else:
            temp = self.head
            while temp.next:
                temp = temp.next
            temp.next = new_node
            new_node.prev = temp
    print("Doubly linked list created successfully.")


# 2. Insert at beginning
def insert_beginning(self):
    data = int(input("Enter data: "))
    new_node = Node(data)
    if self.head is not None:
        new_node.next = self.head
        self.head.prev = new_node
    self.head = new_node
    print("Node inserted at beginning.")


# 3. Insert at end
def insert_end(self):
    data = int(input("Enter data: "))
    new_node = Node(data)
    if self.head is None:
```

```
        self.head = new_node
    else:
        temp = self.head
        while temp.next:
            temp = temp.next
        temp.next = new_node
        new_node.prev = temp
        print("Node inserted at end.")

# 4. Insert at specific index
def insert_index(self):
    index = int(input("Enter index: "))
    data = int(input("Enter data: "))
    new_node = Node(data)
    if index == 0:
        if self.head:
            new_node.next = self.head
            self.head.prev = new_node
        self.head = new_node
        print("Node inserted.")
        return
    temp = self.head
    for i in range(index - 1):
        if temp is None:
            print("Invalid index.")
            return
        temp = temp.next
    if temp is None:
        print("Invalid index.")
        return
    new_node.next = temp.next
    new_node.prev = temp
    if temp.next:
        temp.next.prev = new_node
    temp.next = new_node
    print("Node inserted.")

# 5. Delete by value
def delete_value(self):
    value = int(input("Enter value to delete: "))
    temp = self.head
```

```
    while temp and temp.data != value:
        temp = temp.next
    if temp is None:
        print("Value not found.")
        return
    if temp.prev:
        temp.prev.next = temp.next
    else:
        self.head = temp.next

    if temp.next:
        temp.next.prev = temp.prev

    print("Node deleted.")

# 6. Delete first node
def delete_first(self):
    if self.head is None:
        print("List is empty.")
        return

    self.head = self.head.next

    if self.head:
        self.head.prev = None

    print("First node deleted.")

# 7. Delete last node
def delete_last(self):
    if self.head is None:
        print("List is empty.")
        return

    temp = self.head

    while temp.next:
        temp = temp.next

    if temp.prev:
        temp.prev.next = None
    else:
        self.head = None
```

```
    print("Last node deleted.")

# 8. Count nodes
def count(self):
    count = 0
    temp = self.head

    while temp:
        count += 1
        temp = temp.next

    print("Number of nodes:", count)

# 9. Display / Traverse
def display(self):
    if self.head is None:
        print("List is empty.")
        return
    temp = self.head
    while temp:
        print(temp.data, end=" <-> ")
        temp = temp.next

    print("None")


#Main program
dll = DoublyLinkedList()
while True:
    print("\n--- DOUBLY LINKED LIST ---")
    print("1. Create linked list")
    print("2. Insert at beginning")
    print("3. Insert at end")
    print("4. Insert at specific index")
    print("5. Delete by value")
    print("6. Delete first node")
```

```
    print("7. Delete last node")
    print("8. Count number of nodes")
    print("9. Display / Traverse")
    print("10. Exit")
    choice = int(input("Enter your choice: "))

if choice == 1:
    dll.create()
elif choice == 2:
    dll.insert_beginning()
elif choice == 3:
    dll.insert_end()
elif choice == 4:
    dll.insert_index()
elif choice == 5:
    dll.delete_value()
elif choice == 6:
    dll.delete_first()
elif choice == 7:
    dll.delete_last()
elif choice == 8:
    dll.count()
elif choice == 9:
    dll.display()
elif choice == 10:
    print("Program exited.")
    break
else:
    print("Invalid choice.")
```

OUTPUT:

```
Python 3.13.5 (v3.13.5:6cb20a219a8, Jun 11 2025, 12:23:45) [Clang 16.0.0 (clang-1600.0.26.6)] on darwin
Enter "help" below or click "Help" above for more information.
>>> ========== RESTART: /Users/vamsi/Documents/8SA        CSE25154/LabS    dll.py ==========

--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 1
Enter number of nodes: 5
Enter data for node 1: 11
Enter data for node 2: 81
Enter data for node 3: 18
Enter data for node 4: 25
Enter data for node 5: 2
Doubly linked list created successfully.

--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 2
Enter data: 33
Node inserted at beginning.

--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
```

```
10. Exit
Enter your choice: 3
Enter data: 5
Node inserted at end.

--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 77
Invalid choice.

--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 4
Enter index: 3
Enter data: 24
Node inserted.

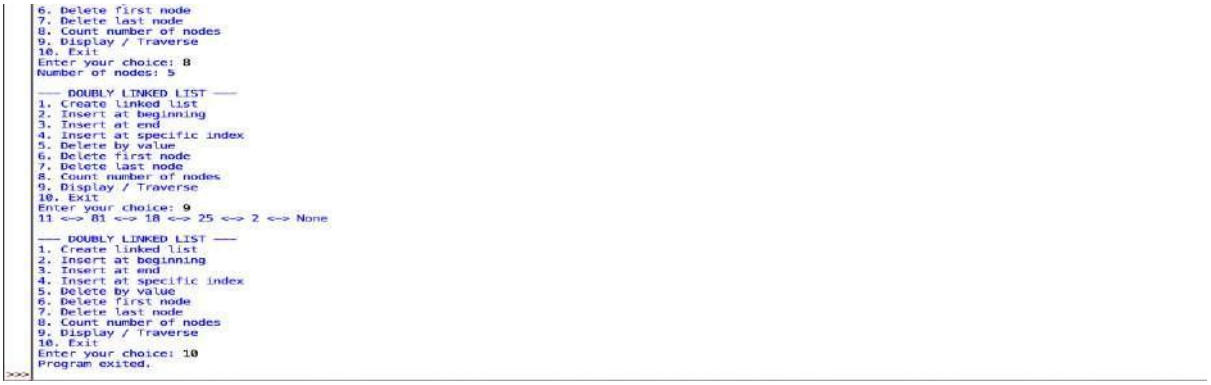
--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 5
Enter value to delete: 24
Node deleted.
```

```
--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 5
Enter value to delete: 34
Value not found.

--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 6
First node deleted.

--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
6. Delete first node
7. Delete last node
8. Count number of nodes
9. Display / Traverse
10. Exit
Enter your choice: 7
Last node deleted.

--- DOUBLY LINKED LIST ---
1. Create linked list
2. Insert at beginning
3. Insert at end
4. Insert at specific index
5. Delete by value
```



EXPLANATION:

- The Node class creates a node containing **data, previous link, and next link**.
- The DoublyLinkedList class maintains the list using the head pointer.
- The create() function creates the doubly linked list by adding the given nodes.
- The insert_beginning() function adds a **new node at the beginning** and updates the previous link.
- The insert_end() function adds a **new node at the end** of the list.
- The insert_index() function inserts a node at a **specified index** and updates both links.
- The delete_value() function searches for a given value and **removes that node**.
- The delete_first() function removes the **first node** and updates the head.
- The delete_last() function removes the **last node** from the list.
- The count() function counts and displays the **number of nodes**.
- The display() function traverses the list and displays the elements using **<->**.
- The main program provides a **menu to select and perform different operations**.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**IndentationError** occurred because the statements were not properly aligned.	Corrected the indentation of all blocks.
2.	**NameError** occurred because new_node, temp, or other variables were misspelled.	Corrected the variable names and spelling.
3.	**AttributeError** occurred when prev or next links were not updated correctly.	Updated the prev and next links properly.
4.	**IndexError** occurred when an invalid index was entered.	Checked whether the index is within the valid range.

C.AIM: Write a Python program to implement a Circular linked list with the above operations.

PROGRAM:

```
class Node:

        def init(self, data):

              self.data = data self.next = None

class CircularLinkedList:

       def init(self):

             self.head = None

# 1. Create linked list
def create(self):
   n = int(input("Enter number of nodes: "))
   for i in range(n):

       data = int(input(f"Enter data for node {i + 1}: "))
       new_node = Node(data)

       if self.head is None:
          self.head = new_node
          new_node.next = self.head
       else:
          temp = self.head

          while temp.next != self.head:
             temp = temp.next

          temp.next = new_node
          new_node.next = self.head

   print("Circular linked list created successfully.")

# 2. Insert at beginning
def insert_beginning(self):
   data = int(input("Enter data: "))
   new_node = Node(data)
```

```
    if self.head is None:
        self.head = new_node
        new_node.next = self.head
    else:
        temp = self.head

        while temp.next != self.head:
            temp = temp.next

        new_node.next = self.head
        temp.next = new_node
        self.head = new_node

    print("Node inserted at beginning.")

# 3. Insert at end
def insert_end(self):
    data = int(input("Enter data: "))
    new_node = Node(data)

    if self.head is None:
        self.head = new_node
        new_node.next = self.head
    else:
        temp = self.head

        while temp.next != self.head:
            temp = temp.next

        temp.next = new_node
        new_node.next = self.head

    print("Node inserted at end.")

# 4. Insert at specific index
def insert_index(self):
    index = int(input("Enter index: "))
    data = int(input("Enter data: "))
    new_node = Node(data)

    if index == 0:
        if self.head is None:
            self.head = new_node
```

```
            new_node.next = self.head
        else:
            temp = self.head

            while temp.next != self.head:
                temp = temp.next

            new_node.next = self.head
            temp.next = new_node
            self.head = new_node

        print("Node inserted.")
        return

    if self.head is None:
        print("Invalid index.")
        return

    temp = self.head

    for i in range(index - 1):
        temp = temp.next

        if temp == self.head:
            print("Invalid index.")
            return

    new_node.next = temp.next
    temp.next = new_node

    print("Node inserted.")

# 5. Delete by value
def delete_value(self):
    value = int(input("Enter value to delete: "))

    if self.head is None:
        print("List is empty.")
        return

    # If head contains the value
    if self.head.data == value:
```

```
        if self.head.next == self.head:
            self.head = None
        else:
            temp = self.head

            while temp.next != self.head:
                temp = temp.next

            temp.next = self.head.next
            self.head = self.head.next

        print("Node deleted.")
        return

    temp = self.head

    while temp.next != self.head and temp.next.data != value:
        temp = temp.next

    if temp.next == self.head:
        print("Value not found.")
    else:
        temp.next = temp.next.next
        print("Node deleted.")

# 6. Delete first node
def delete_first(self):
    if self.head is None:
        print("List is empty.")
        return

    if self.head.next == self.head:
        self.head = None
    else:
        temp = self.head

        while temp.next != self.head:
            temp = temp.next

        temp.next = self.head.next
        self.head = self.head.next
```

```
    print("First node deleted.")

# 7. Delete last node
def delete_last(self):
    if self.head is None:
        print("List is empty.")
        return

    if self.head.next == self.head:
        self.head = None
    else:
        temp = self.head

        while temp.next.next != self.head:
            temp = temp.next

        temp.next = self.head

    print("Last node deleted.")

# 8. Count nodes
def count(self):
    if self.head is None:
        print("Number of nodes: 0")
        return

    count = 0
    temp = self.head

    while True:
        count += 1
        temp = temp.next

        if temp == self.head:
            break

    print("Number of nodes:", count)
# 9. Display / Traverse
def display(self):
    if self.head is None:
        print("List is empty.")
        return
```

```
    temp = self.head

    while True:
        print(temp.data, end=" -> ")
        temp = temp.next

        if temp == self.head:
            break

    print("HEAD")
 #Main program

cll = CircularLinkedList()

while True:

    print("\n--- CIRCULAR LINKED LIST ---")

    print("1. Create linked list")

    print("2. Insert at beginning")

    print("3. Insert at end")

    print("4. Insert at specific index")

    print("5. Delete by value")

    print("6. Delete first node")

    print("7. Delete last node")

    print("8. Count number of nodes")

    print("9. Display / Traverse")

    print("10. Exit")

choice = int(input("Enter your choice: "))

if choice == 1:
    cll.create()
elif choice == 2:
    cll.insert_beginning()
elif choice == 3:
```

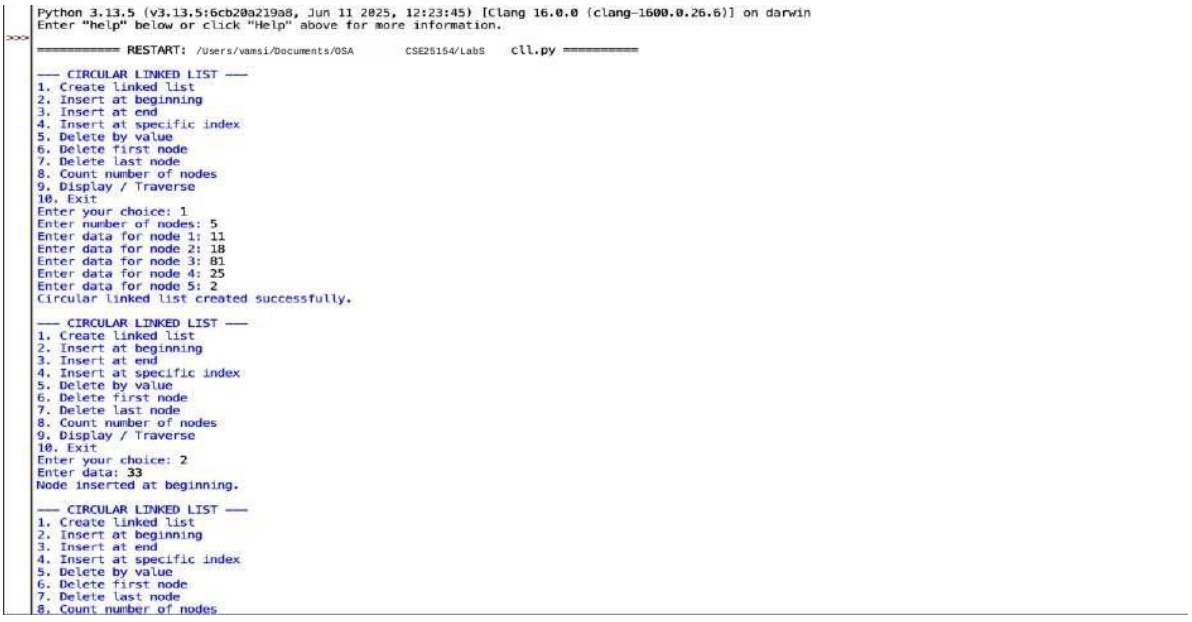
```
        cll.insert_end()
    elif choice == 4:
        cll.insert_index()
    elif choice == 5:
        cll.delete_value()
    elif choice == 6:
        cll.delete_first()
    elif choice == 7:
        cll.delete_last()
    elif choice == 8:
        cll.count()

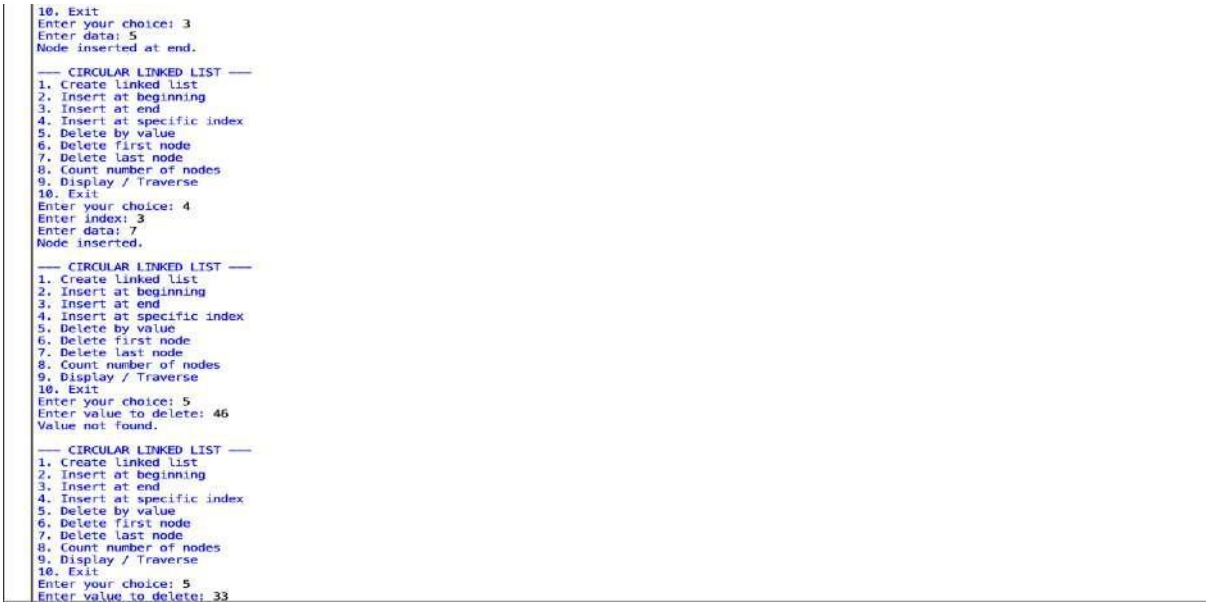
    elif choice == 9:
        cll.display()

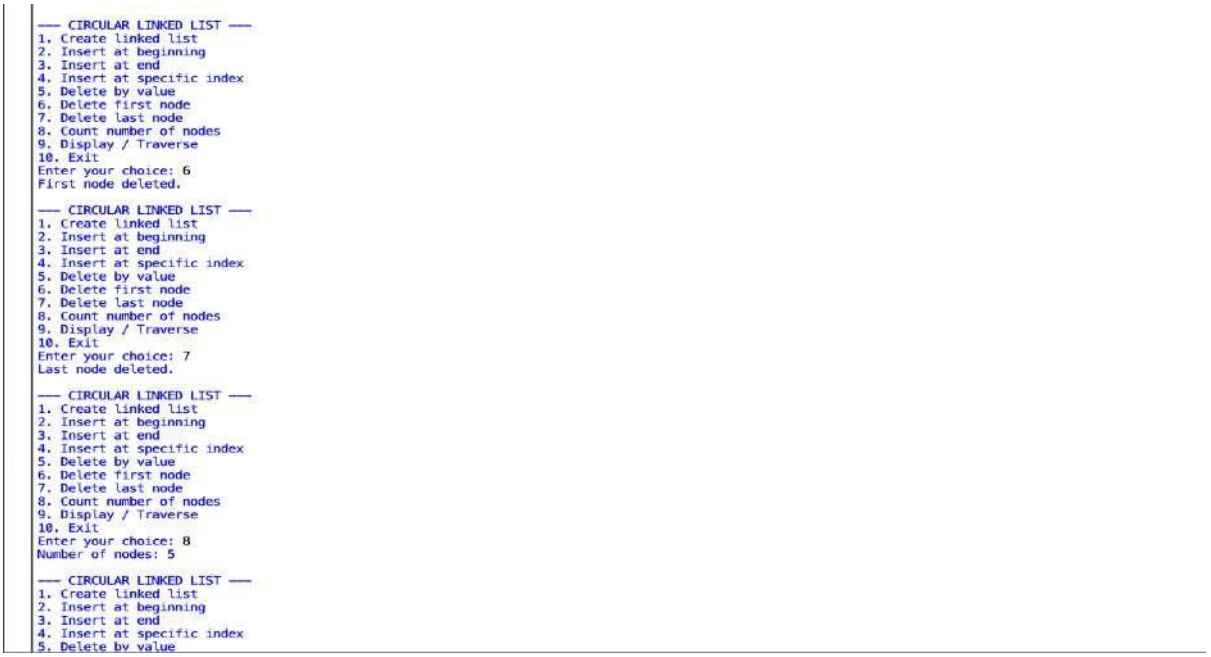
    elif choice == 10:
        print("Program exited.")
        break

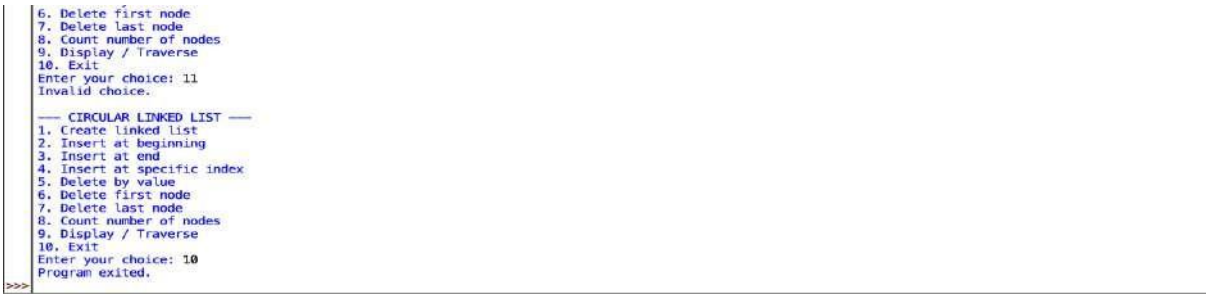
    else:
        print("Invalid choice.")
```

OUTPUT:









EXPLANATION:

- The Node class creates a node containing **data and a link to the next node**.
- The CircularLinkedList class maintains the list using the head pointer.

- The create() function creates the circular linked list and connects the **last node back to the head**.
- The insert_beginning() function adds a **new node at the beginning** and updates the circular connection.
- The insert_end() function adds a **new node at the end** and connects it back to the head.
- The insert_index() function inserts a node at a **specified index**.
- The delete_value() function searches for a value and **deletes the corresponding node**.
- The delete_first() function removes the **first node** and updates the head.
- The delete_last() function removes the **last node** while maintaining the circular connection.
- The count() function counts and displays the **total number of nodes**.
- The display() function traverses the list and displays the elements until it reaches the **head again**.
- The main program provides a **menu to perform different circular linked-list operations**.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**IndentationError** occurred because the statements were not properly aligned.	Corrected the indentation of all code blocks.
2.	**NameError** occurred because variables like temp or new_node were misspelled.	Corrected the variable names and spelling.
3.	**AttributeError** occurred when the next link was not updated properly.	Updated the next links correctly to maintain the circular connection.
4.	**IndexError** occurred when an invalid index was entered.	Checked whether the index is within the valid range.
5.	**Value not found** occurred when the entered value was not present in the list.	Searched the list and displayed **"Value not found."**

Week-6

A.AIM: Write a Python program to implement stack operations such as Push, Pop, Peek, Display using an array

PROGRAM:

```
stack = []

# Push operation

def push():

    value = int(input("Enter element: "))

    stack.append(value)

    print("Element pushed.")

# Pop operation

def pop():

    if len(stack) == 0:

        print("Stack is empty.")

    else:

        print("Popped element:", stack.pop())

# Peek operation

def peek():

    if len(stack) == 0:

        print("Stack is empty.")

    else:

        print("Top element:", stack[-1])

# Display operation

def display():
```

```
    if len(stack) == 0:
        print("Stack is empty.")
    else:
        print("Stack elements:")
        for i in range(len(stack) - 1, -1, -1):
            print(stack[i])
# Main program
while True:
    print("\n--- STACK USING ARRAY ---")
    print("1. Push")
    print("2. Pop")
    print("3. Peek")
    print("4. Display")
    print("5. Exit")
    choice = int(input("Enter your choice: "))
    if choice == 1:
        push()
    elif choice == 2:
        pop()
    elif choice == 3:
        peek()
    elif choice == 4:
        display()
    elif choice == 5:
        print("Program exited.")
```

```
        break
    else:
        print("Invalid choice.")
```

OUTPUT:



```
Python 3.13.5 (v3.13.5:6cb20a219a8, Jun 11 2025, 12:23:45) [Clang 16.0.0 (clang-1600.0.26.6)] on darwin
Enter "help" below or click "Help" above for more information.
>>> ========== RESTART: /Users/vamsi/Documents/@SA        CSE25154/Lab6.py     ==========

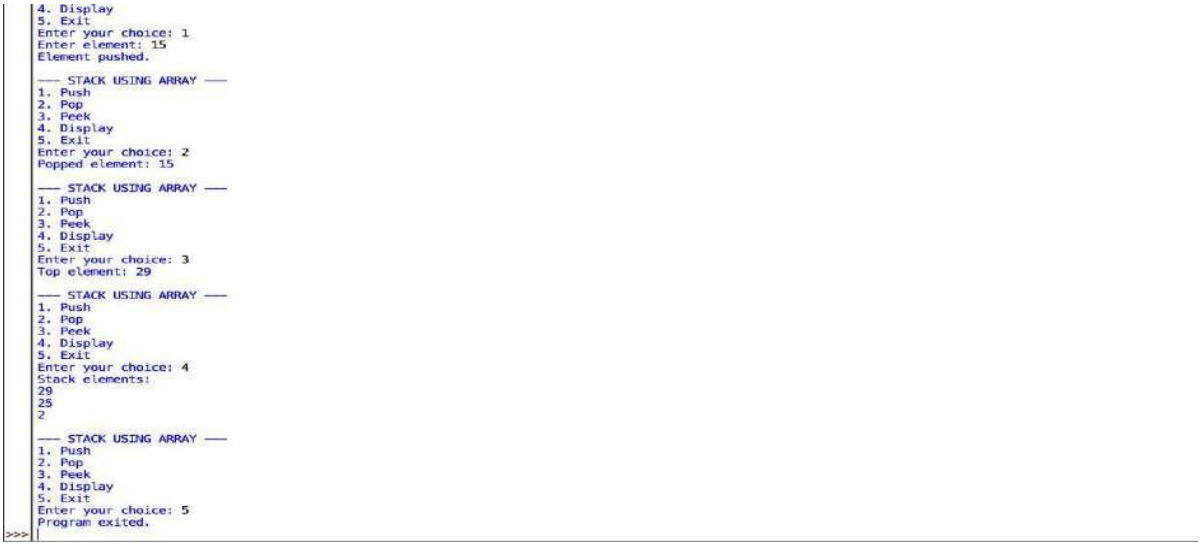
--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 8
Invalid choice.

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 2
Element pushed.

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 25
Element pushed.

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 29
Element pushed.

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
```



```
4. Display
5. Exit
Enter your choice: 1
Enter element: 15
Element pushed.

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 2
Popped element: 15

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 3
Top element: 29

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack elements:
29
25
2

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 5
Program exited.
>>>
```

EXPLANATION:

- The stack list is used to store the **stack elements**.

- The push() function adds an element to the **top of the stack**.
- The pop() function removes and returns the **top element**.
- The peek() function displays the **top element without removing it**.
- The display() function prints all elements from **top to bottom**.
- len(stack) == 0 checks whether the stack is **empty**.
- The main program provides a **menu** to perform different stack operations.
- Stack follows the **LIFO (Last In, First Out)** principle.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**ValueError** occurred when a non-integer value was entered.	Entered only valid integer values.
2.	**NameError** occurred because stack or a function name was misspelled.	Corrected the variable and function names.
3.	**Index Error** occurred when trying to access the top element of an empty stack.	Checked whether the stack is empty before accessing an element.
4.	**Empty stack error** occurred when Pop or Peek was performed without any elements.	Checked len(stack) == 0 before performing Pop or Peek.
5.	**SyntaxError** occurred because of incorrect Python syntax.	Checked and corrected the syntax of the program.

B.AIM: Write a Python program to implement stack operations such as Push, Pop, Peek, Display using a linked list

PROGRAM:

```
class Node:

    def init(self, data):

        self.data = data self.next = None

class Stack:

     def init(self):

        self.top = None

# Push operation
def push(self):
   value = int(input("Enter element: "))

   new_node = Node(value)
   new_node.next = self.top
   self.top = new_node

   print("Element pushed.")

# Pop operation
def pop(self):
   if self.top is None:
      print("Stack is empty.")
   else:
      print("Popped element:", self.top.data)
      self.top = self.top.next

# Peek operation
def peek(self):
   if self.top is None:
      print("Stack is empty.")
   else:
      print("Top element:", self.top.data)

# Display operation
def display(self):
```

```
    if self.top is None:
        print("Stack is empty.")
    else:
        temp = self.top

        print("Stack elements:")
        while temp:
            print(temp.data)
            temp = temp.next


#Main program

stack = Stack()

while True:

    print("\n--- STACK USING LINKED LIST ---")

    print("1. Push")

    print("2. Pop")

    print("3. Peek")

    print("4. Display")

    print("5. Exit")

choice = int(input("Enter your choice: "))

if choice == 1:
    stack.push()

elif choice == 2:
    stack.pop()

elif choice == 3:
    stack.peek()

elif choice == 4:
    stack.display()

elif choice == 5:
    print("Program exited.")
```

```
    break

else:
    print("Invalid choice.")
```

OUTPUT:



```
Python 3.13.5 (v3.13.5:6cb20a219a8, Jun 11 2025, 12:23:45) [Clang 16.0.0 (clang-1600.0.26.6)] on darwin
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: /Users/vamsi/Documents/DSA        CSE25154/Lab6(b)     ll.py ========

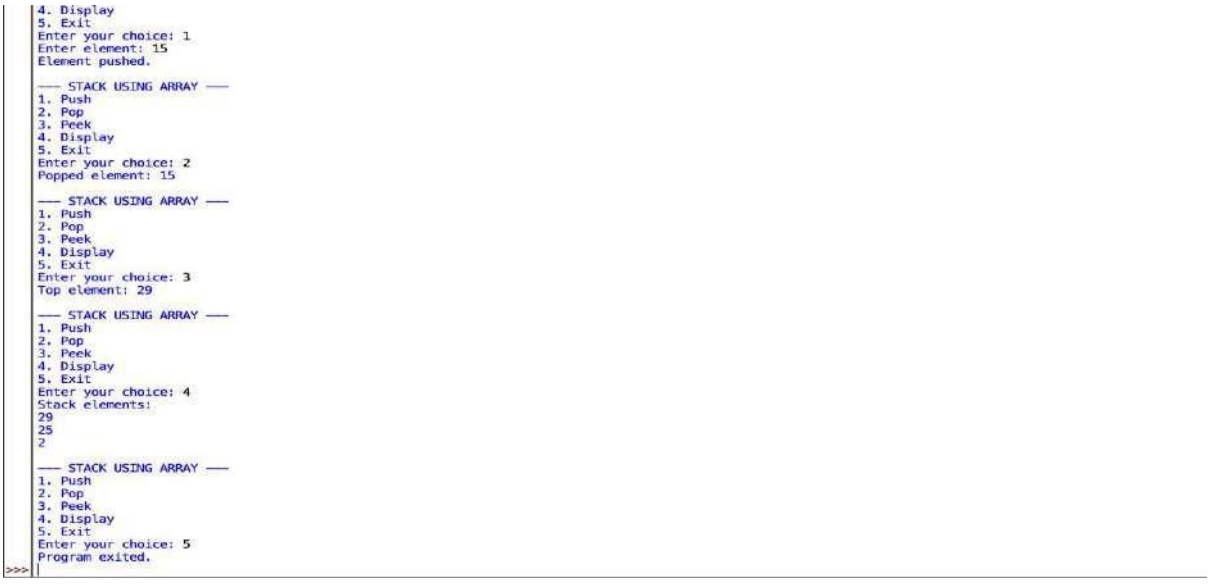
--- STACK USING LINKED LIST ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 2
Element pushed.

--- STACK USING LINKED LIST ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 25
Element pushed.

--- STACK USING LINKED LIST ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 29
Element pushed.

--- STACK USING LINKED LIST ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 15
Element pushed.

--- STACK USING LINKED LIST ---
1. Push
2. Pop
3. Peek
```



```
4. Display
5. Exit
Enter your choice: 1
Enter element: 15
Element pushed.

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 2
Popped element: 15

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 3
Top element: 29

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack elements:
29
25
2

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 5
Program exited.
>>>
```

EXPLANATION:

- The Node class creates a node containing **data and a link to the next node**.

- The Stack class uses top to keep track of the **top element**.
- The push() function adds a new node at the **top of the stack**.
- The pop() function removes the **top node** from the stack.
- The peek() function displays the **top element without removing it**.
- The display() function traverses the linked list and displays all elements.
- self.top is None checks whether the stack is **empty**.
- The main program provides a **menu** for performing stack operations.
- The stack follows the **LIFO (Last In, First Out)** principle.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**NameError** occurred because top, temp, or new_node was misspelled.	Corrected the variable names and their spelling.
2.	**AttributeError** occurred when next was accessed incorrectly.	Correctly used self.top and temp.next.
3.	**ValueError** occurred when a non-integer value was entered.	Entered only valid integer values.
4.	**Empty stack error** occurred when Pop or Peek was performed without any elements.	Checked self.top is None before performing the operation.
5.	**IndentationError** occurred because the statements inside the functions were not properly aligned.	Corrected the indentation of all code blocks.

Week-7

A.AIM: Write a Python program to implement a queue and its operations such as Enqueue, Dequeue, Peek, Display with a menu-driven approach. Implement using array

PROGRAM:

```
queue = []

# Enqueue operation

def enqueue():

    value = int(input("Enter element: "))

    queue.append(value)

    print("Element inserted.")

 # Dequeue operation

def dequeue():

    if len(queue) == 0:

        print("Queue is empty.")

    else:

        print("Deleted element:", queue.pop(0))

 # Peek operation

def peek():

    if len(queue) == 0:

        print("Queue is empty.")

    else:

        print("Front element:", queue[0])

 # Display operation

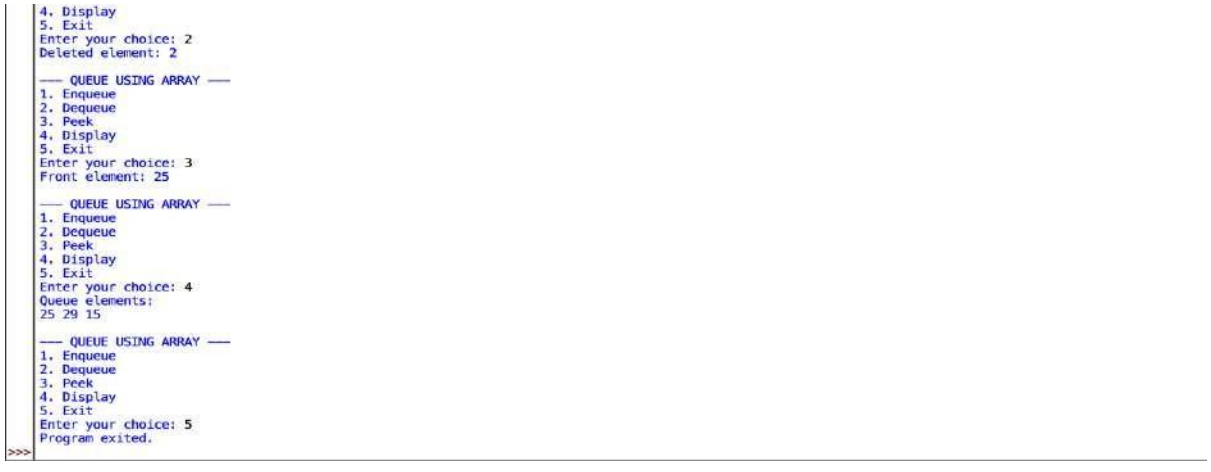
def display():
```

```
    if len(queue) == 0:
        print("Queue is empty.")
    else:
        print("Queue elements:")
        for element in queue:
            print(element, end=" ")
        print()
# Main program
while True:
    print("\n--- QUEUE USING ARRAY ---")
    print("1. Enqueue")
    print("2. Dequeue")
    print("3. Peek")
    print("4. Display")
    print("5. Exit")
    choice = int(input("Enter your choice: "))
    if choice == 1:
        enqueue()
    elif choice == 2:
        dequeue()
    elif choice == 3:
        peek()
    elif choice == 4:
        display()
    elif choice == 5:
```

```
    print("Program exited.")
    break
else:
    print("Invalid choice.")
```

OUTPUT:





EXPLANATION:

- The queue list is used to store the **queue elements**.
- The enqueue() function adds an element at the **rear** of the queue.

- The dequeue() function removes an element from the **front** of the queue.
- The peek() function displays the **front element** without removing it.
- The display() function prints all the elements in the queue.
- len(queue) == 0 checks whether the queue is **empty**.
- The main program provides a **menu** to select different queue operations.
- Queue follows the **FIFO (First In, First Out)** principle.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**IndentationError** occurred because the statements inside the functions were not properly aligned.	Corrected the indentation of all code blocks.
2.	**ValueError** occurred when a non-integer value was entered.	Entered only valid integer values.
3.	**NameError** occurred because queue or a function name was misspelled.	**NameError** occurred because queue or a function name was misspelled.
4.	**IndexError** occurred when trying to access an element from an empty queue.	Checked whether the queue is empty before accessing elements.
5.	**Empty queue error** occurred when Dequeue or Peek was performed without elements.	Checked len(queue) == 0 before performing the operation.

B.AIM: Write a Python program to implement a queue and its operations such as Enqueue, Dequeue, Peek, Display with a menu-driven approach. Implement using Linked list data.

PROGRAM:

```
class Node:

    def init(self, data):

        self.data = data self.next = None

class Queue:

   def init(self):

        self.front = None self.rear = None

# Enqueue operation
def enqueue(self):
   value = int(input("Enter element: "))
   new_node = Node(value)

   if self.rear is None:
      self.front = self.rear = new_node
   else:
      self.rear.next = new_node
      self.rear = new_node

   print("Element inserted.")

# Dequeue operation
def dequeue(self):
   if self.front is None:
      print("Queue is empty.")
   else:
      print("Deleted element:", self.front.data)
      self.front = self.front.next

      if self.front is None:
         self.rear = None

# Peek operation
def peek(self):
```

```
    if self.front is None:
        print("Queue is empty.")
    else:
        print("Front element:", self.front.data)

# Display operation
def display(self):
    if self.front is None:
        print("Queue is empty.")
    else:
        temp = self.front

        print("Queue elements:")
        while temp:
            print(temp.data, end=" ")
            temp = temp.next

        print()
 #Main program
queue = Queue()

while True:

    print("\n--- QUEUE USING LINKED LIST ---")

    print("1. Enqueue")

    print("2. Dequeue")

    print("3. Peek")

    print("4. Display")

    print("5. Exit")

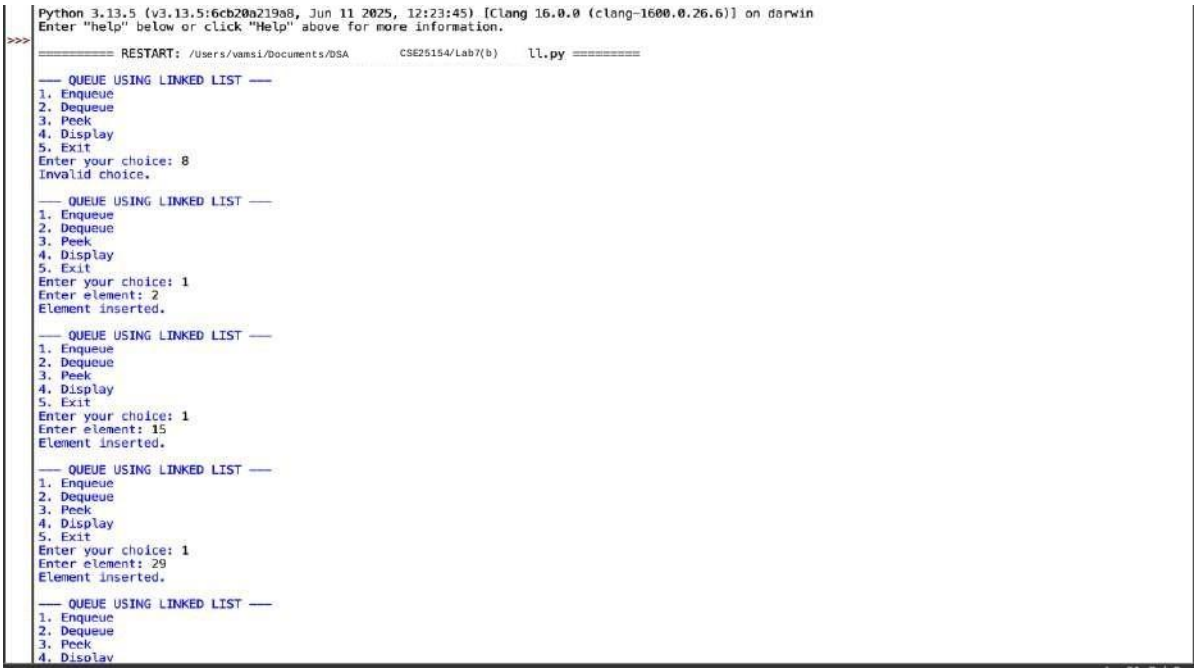
choice = int(input("Enter your choice: "))

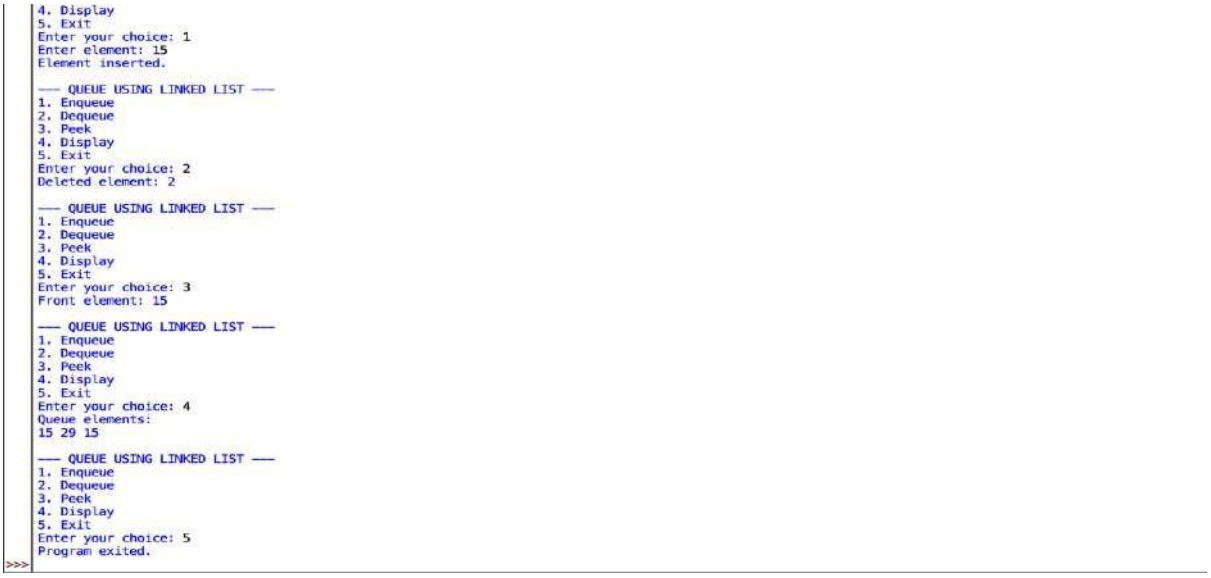
if choice == 1:
    queue.enqueue()
elif choice == 2:
    queue.dequeue()

elif choice == 3:
    queue.peek()
elif choice == 4:
```

```
    queue.display()
elif choice == 5:
    print("Program exited.")
    break
else:
    print("Invalid choice.")
```

OUTPUT:





EXPLANATION:

- The Node class creates a node containing **data and a link to the next node**.
- The Queue class uses front and rear to maintain the queue.

- The enqueue() function adds a new node at the **rear**.
- The dequeue() function removes a node from the **front**.
- The peek() function displays the **front element** without removing it.
- The display() function traverses the linked list and displays all elements.
- The queue follows the **FIFO (First In, First Out)** principle.
- The main program provides a **menu-driven approach** for performing the operations.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**NameError** occurred because front, rear, or new_node was misspelled.	Corrected the variable names and their spelling.
2.	**AttributeError** occurred when next was accessed incorrectly.	Used self.front, self.rear, and temp.next correctly.
3.	**ValueError** occurred when a non-integer value was entered.	Entered only valid integer values.
4.	**Empty queue error** occurred when Dequeue or Peek was performed without elements.	Checked self.front is None before performing the operation.
5.	**SyntaxError** occurred due to incorrect Python syntax.	Checked and corrected the syntax of the program.

C.AIM: Write a Python program to implement a queue and its operations such as Enqueue, Dequeue, Peek, Display with a menu-driven approach. Implement using Circular Queue array structures

PROGRAM:

```
class CircularQueue:

      def init(self, size):

            self.size = size self.queue = [None] * size self.front = -1 self.rear = -1

# Enqueue
def enqueue(self):
   value = int(input("Enter element: "))

  if (self.rear + 1) % self.size == self.front:
      print("Queue is full.")
      return

   if self.front == -1:
      self.front = 0

   self.rear = (self.rear + 1) % self.size
   self.queue[self.rear] = value

   print("Element inserted.")
# Dequeue
def dequeue(self):
   if self.front == -1:
      print("Queue is empty.")
      return

   print("Deleted element:", self.queue[self.front])

   if self.front == self.rear:
      self.front = -1
      self.rear = -1
   else:
      self.front = (self.front + 1) % self.size
# Peek
def peek(self):
   if self.front == -1:
      print("Queue is empty.")
   else:
      print("Front element:", self.queue[self.front])
```

```
# Display
def display(self):
    if self.front == -1:
        print("Queue is empty.")
        return

    print("Queue elements:")
    i = self.front

    while True:
        print(self.queue[i], end=" ")

        if i == self.rear:
            break

        i = (i + 1) % self.size

    print()
#Main program

size = int(input("Enter queue size: "))

queue = CircularQueue(size)

while True:

    print("\n--- CIRCULAR QUEUE ---")

    print("1. Enqueue")

    print("2. Dequeue")

    print("3. Peek")

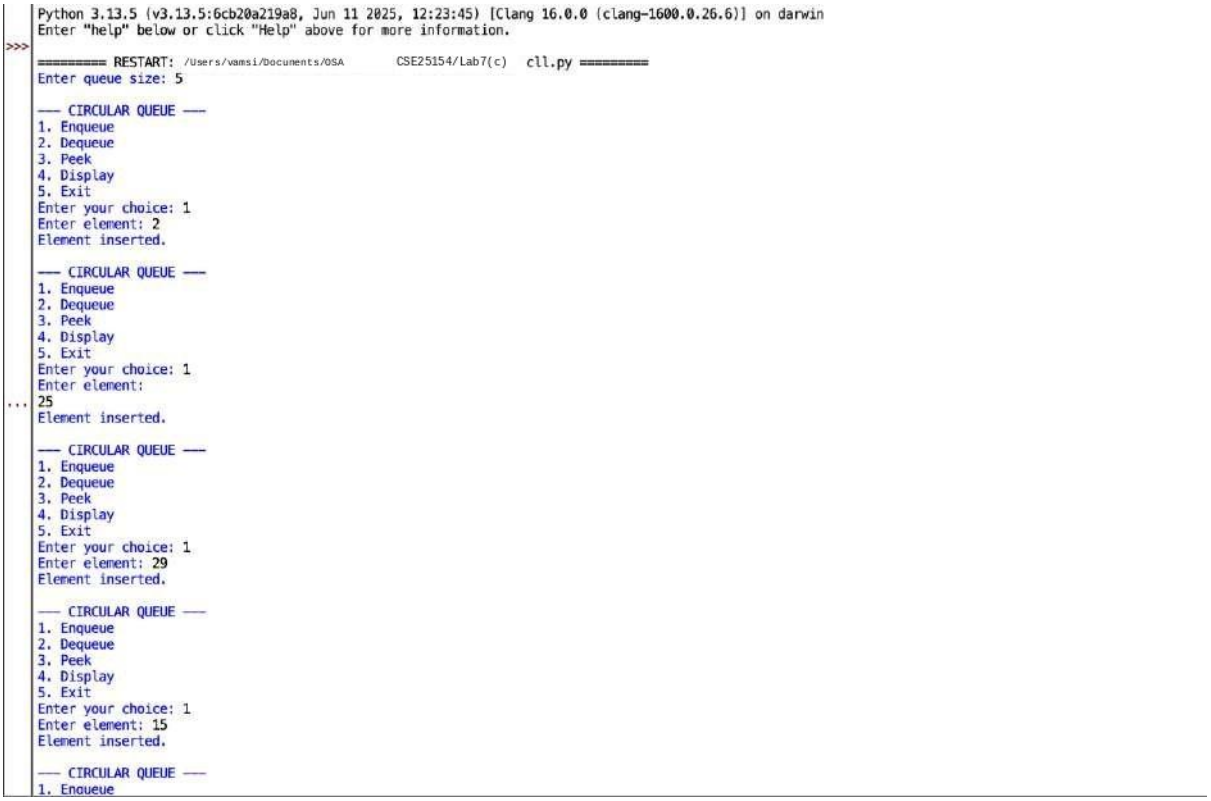
    print("4. Display")

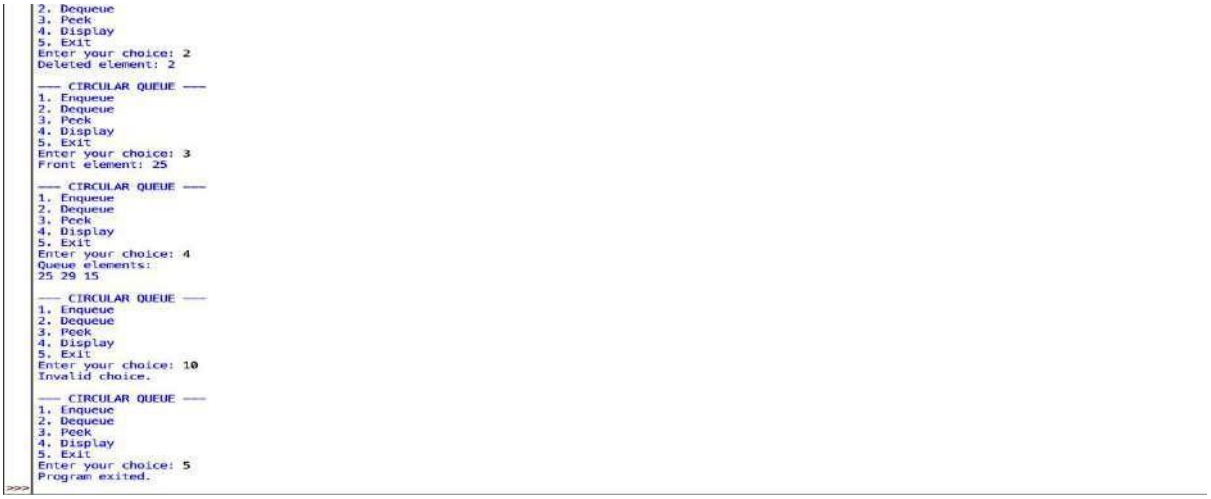
    print("5. Exit")

choice = int(input("Enter your choice: "))
if choice == 1:
    queue.enqueue()
elif choice == 2:
    queue.dequeue()
elif choice == 3:
    queue.peek()
```

```
elif choice == 4:
    queue.display()
elif choice == 5:
    print("Program exited.")
    break
else:
    print("Invalid choice.")
```

OUTPUT:





EXPLANATION:

- The CircularQueue class creates a **queue using an array**.

- front represents the position of the **first element**, while rear represents the **last element**.
- enqueue() adds a new element at the **rear**.
- dequeue() removes an element from the **front**.
- peek() displays the **front element** without deleting it.
- display() prints all the elements from **front to rear**.
- The % size operation makes the rear and front **wrap around the array**.
- The main program provides a **menu-driven interface**.
- The queue follows the **FIFO (First In, First Out)** principle.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**IndentationError** occurred because the statements were not properly aligned.	Corrected the indentation of all code blocks.
2.	**ValueError** occurred when a non-integer value was entered.	Entered only valid integer values.
3.	**NameError** occurred because front, rear, or queue was misspelled.	Corrected the variable names and spelling.
4.	**IndexError** occurred when an invalid array position was accessed.	Used proper circular indexing with % size.
5.	**Queue full error** occurred when an element was inserted into a full queue.	Checked the full condition before enqueueing.

D.AIM: Write a Python program to implement a queue and its operations such as Enqueue, Dequeue, Peek, Display with a menu-driven approach. Implement using Circular Queue Linked list structures.

PROGRAM:

```
class Node:

    def init(self, data):

        self.data = data self.next = None

class CircularQueue:

    def init(self):

        self.front = None self.rear = None

# Enqueue
def enqueue(self):

    value = int(input("Enter element: "))
    new_node = Node(value)

    if self.front is None:
        self.front = self.rear = new_node
        new_node.next = self.front
    else:
        new_node.next = self.front
        self.rear.next = new_node
        self.rear = new_node

    print("Element inserted.")
# Dequeue
def dequeue(self):
    if self.front is None:
        print("Queue is empty.")
        return

    print("Deleted element:", self.front.data)

    if self.front == self.rear:
        self.front = None
        self.rear = None
    else:
```

```
        self.front = self.front.next
        self.rear.next = self.front

# Peek
def peek(self):
    if self.front is None:
        print("Queue is empty.")
    else:
        print("Front element:", self.front.data)

# Display
def display(self):
    if self.front is None:
        print("Queue is empty.")
        return

    temp = self.front

    print("Queue elements:")

    while True:
        print(temp.data, end=" ")

        temp = temp.next

       if temp == self.front:
            break

    print()


#Main program

queue = CircularQueue()

while True:

    print("\n--- CIRCULAR QUEUE USING LINKED LIST ---")

    print("1. Enqueue")

    print("2. Dequeue")

    print("3. Peek")
```

```
    print("4. Display")

    print("5. Exit")

choice = int(input("Enter your choice: "))
if choice == 1:
    queue.enqueue()
elif choice == 2:
    queue.dequeue()
elif choice == 3:
    queue.peek()
elif choice == 4:
    queue.display()
elif choice == 5:
    print("Program exited.")
    break
else:
    print("Invalid choice.")
```

OUTPUT:

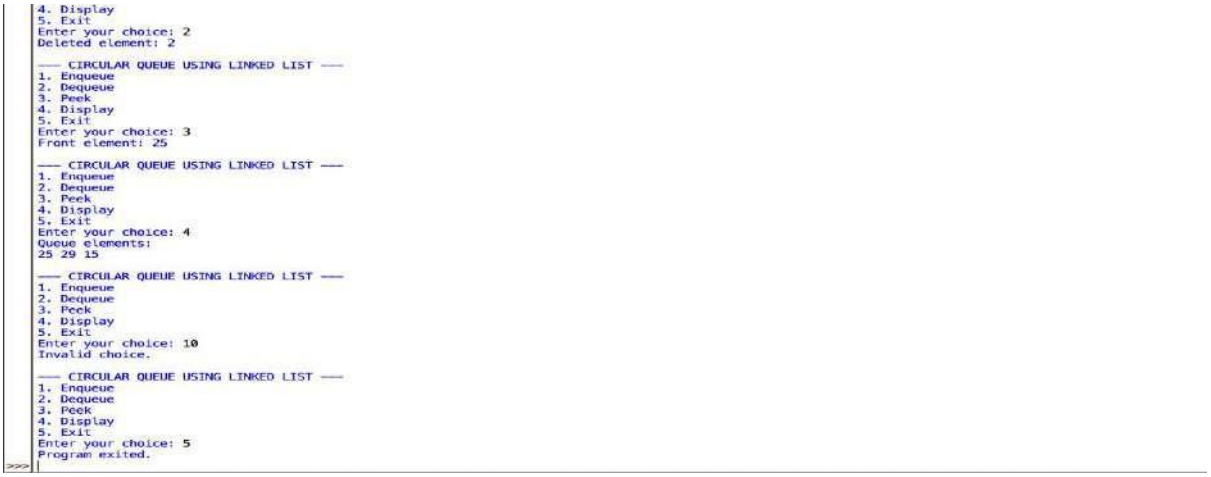
```
Python 3.13.5 (v3.13.5:6cb20e219a8, Jun 11 2025, 12:23:45) [Clang 16.0.0 (clang-1600.0.26.6)] on darwin
Enter "help" below or click "Help" above for more information.
>>>
[1] ========= RESTART: /Users/vamsi/Documents/DSA  CSE25154/Lab7(d)  cll ll.py =======

--- CIRCULAR QUEUE USING LINKED LIST ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter clement: 3
Element inserted.

--- CIRCULAR QUEUE USING LINKED LIST ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter clement: 25
Element inserted.

--- CIRCULAR QUEUE USING LINKED LIST ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter clement: 20
Element inserted.

--- CIRCULAR QUEUE USING LINKED LIST ---
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter clement: 15
Element inserted.
```



EXPLANATION:

- The Node class creates a node containing **data and a next link**.
- The CircularQueue class maintains the queue using front and rear.
- The enqueue() function adds a node at the **rear** and connects it back to the front.
- The dequeue() function removes the **front node**.
- The peek() function displays the **front element** without removing it.
- The display() function traverses the circular linked list until it reaches the **front again**.
- front and rear are set to None when the queue becomes empty.
- The main program provides a **menu-driven approach**.
- The circular queue follows the **FIFO (First In, First Out)** principle.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**NameError** occurred because front, rear, or new_node was misspelled.	Corrected the variable names and spelling.
2.	**AttributeError** occurred when the next link was not accessed correctly.	Used self.front, self.rear, and temp.next correctly.
3.	**ValueError** occurred when a non-integer value was entered.	Entered only valid integer values.
4.	**Empty queue error** occurred when Dequeue or Peek was performed on an empty queue.	Checked self.front is None before performing the operation.
5.	**Circular link error** occurred when the rear node was not connected back to the front node.	Updated self.rear.next = self.front after insertion and deletion.

Week-8

A.AIM: Write a Python program to implement a binary tree with its operations using an arrays

PROGRAM:

```
class BinaryTreeArray:

      def init(self, size):

           self.tree = [None] * size

           self.size = size

# Insert root
def set_root(self, data):

   if self.size > 0:
      self.tree[0] = data
   else:
      print("Tree is empty!")

# Set left child
def set_left(self, parent_index, data):
   if parent_index < 0 or parent_index >= self.size:
      print("Invalid parent index!")
      return

   if self.tree[parent_index] is None:
      print("Parent node does not exist!")
      return

   child_index = 2 * parent_index + 1

   if child_index < self.size:
      self.tree[child_index] = data
   else:
      print("Index out of range!")

# Set right child
def set_right(self, parent_index, data):
   if parent_index < 0 or parent_index >= self.size:
```

```
            print("Invalid parent index!")
            return

        if self.tree[parent_index] is None:
            print("Parent node does not exist!")
            return

        child_index = 2 * parent_index + 2

        if child_index < self.size:
            self.tree[child_index] = data
        else:
            print("Index out of range!")

    # Preorder Traversal
    def preorder(self, index=0):
        if index >= self.size or self.tree[index] is None:
            return

        print(self.tree[index], end=" ")
        self.preorder(2 * index + 1)
        self.preorder(2 * index + 2)

    # Inorder Traversal
    def inorder(self, index=0):
        if index >= self.size or self.tree[index] is None:
            return

        self.inorder(2 * index + 1)
        print(self.tree[index], end=" ")
        self.inorder(2 * index + 2)

    # Postorder Traversal
    def postorder(self, index=0):
        if index >= self.size or self.tree[index] is None:
            return

        self.postorder(2 * index + 1)
        self.postorder(2 * index + 2)
        print(self.tree[index], end=" ")

    # Level-order Traversal
    def level_order(self):
```

```
    for value in self.tree:
        if value is not None:
            print(value, end=" ")

# Display Array
def display(self):
    print("\nArray Representation:")
    for i in range(self.size):
        print(f"Index {i}: {self.tree[i]}")
```

Main Program

```
tree = BinaryTreeArray(7)

tree.set_root('A')

tree.set_left(0, 'B') tree.set_right(0, 'C')

tree.set_left(1, 'D') tree.set_right(1, 'E')

tree.set_right(2, 'F')

tree.display()

print("\nPreorder Traversal:", end=" ") tree.preorder()

print("\nInorder Traversal:", end=" ") tree.inorder()

print("\nPostorder Traversal:", end=" ") tree.postorder()

print("\nLevel-order Traversal:", end=" ") tree.level_order()

print()
```

OUTPUT:

```
========= RESTART: /Users/vamsi/Documents/DSA                CSE25154/Lab8(a)btarray.py =========

Array Representation:
Index 0: A
Index 1: B
Index 2: C
Index 3: D
Index 4: E
Index 5: None
Index 6: F

Preorder Traversal: A B D E C F
Inorder Traversal: D B E A C F
Postorder Traversal: D E B F C A
Level-order Traversal: A B C D E F
>>
```

EXPLANATION:

- The BinaryTreeArray class represents a binary tree using an array.
- The __init__() function initializes an array of the given size with None values.
- The set_root() function inserts the root node at index 0.
- The set_left() function inserts the left child of a parent node using the formula 2 * parent_index + 1.
- The set_right() function inserts the right child of a parent node using the formula 2 * parent_index + 2.
- The preorder() function traverses the tree in the order Root → Left → Right.
- The inorder() function traverses the tree in the order Left → Root → Right.
- The postorder() function traverses the tree in the order Left → Right → Root.
- The level_order() function displays the nodes level by level by traversing the array.
- The display() function displays each array index and its corresponding node value.
- The main program creates a binary tree of size 7 and inserts nodes A, B, C, D, E, F.
- The binary tree follows the parent-child relationship, where each node can have at most two children: a left child and a right child.
- The program demonstrates array representation and different tree traversal techniques.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**IndexError** may occur if an invalid index is accessed in the tree array.	Checked the index range before accessing the array.
2.	**Incorrect child index** occurred when the left or right child position was calculated incorrectly.	Used 2 * parent_index + 1 for the left child and 2 * parent_index + 2 for the right child.
3.	**Parent node error** occurred when trying to insert a child for a parent that does not exist.	Checked whether self.tree[parent_index] is None before insertion.
4.	Index out of range error occurred when the child index exceeded the array size.	Checked whether child_index < self.size before inserting the child.
5.	Incorrect traversal output occurred when the preorder, inorder, or postorder sequence was not followed correctly.	Used the correct traversal orders: Root–Left–Right, Left–Root–Right, and Left–Right–Root.

B.AIM: Write a Python program to implement a binary tree with its operations using an Linked list

PROGRAM:

```
#Binary Tree Using Linked List

class Node: def init(self, data): self.data = data self.left = None self.right = None

class BinaryTree: def init(self): self.root = None

# Insert root
def set_root(self, data):
    if self.root is None:
        self.root = Node(data)
    else:
        print("Root already exists!")

# Set left child
def set_left(self, parent, data):
    node = self.search(parent)

    if node is None:
        print("Parent node not found!")
    elif node.left is not None:
        print("Left child already exists!")
    else:
        node.left = Node(data)

# Set right child
def set_right(self, parent, data):
    node = self.search(parent)

    if node is None:
        print("Parent node not found!")
    elif node.right is not None:
        print("Right child already exists!")
    else:
        node.right = Node(data)

# Search for a node by its value
def search(self, data):
    if self.root is None:
```

```
        return None

    queue = [self.root]

    while queue:
        node = queue.pop(0)

        if node.data == data:
            return node

        if node.left is not None:
            queue.append(node.left)

        if node.right is not None:
            queue.append(node.right)

    return None

# Preorder: Root, Left, Right
def preorder(self, node):
    if node is not None:
        print(node.data, end=" ")
        self.preorder(node.left)
        self.preorder(node.right)

# Inorder: Left, Root, Right
def inorder(self, node):
    if node is not None:
        self.inorder(node.left)
        print(node.data, end=" ")
        self.inorder(node.right)

# Postorder: Left, Right, Root
def postorder(self, node):
    if node is not None:
        self.postorder(node.left)
        self.postorder(node.right)
        print(node.data, end=" ")

# Level-order traversal
def level_order(self):
    if self.root is None:
        return
```

```
    queue = [self.root]

    while queue:
        node = queue.pop(0)
        print(node.data, end=" ")

        if node.left is not None:
            queue.append(node.left)

        if node.right is not None:
            queue.append(node.right)

# Display tree using preorder
def display(self):
    if self.root is None:
        print("Tree is empty!")
    else:
        self.preorder(self.root)
        print()
```

Main Program

```
tree = BinaryTree()

tree.set_root('A')

tree.set_left('A', 'B') tree.set_right('A', 'C')

tree.set_left('B', 'D') tree.set_right('B', 'E')

tree.set_right('C', 'F')

print("Binary Tree (Preorder): ", end="") tree.display()

print("Preorder Traversal: ", end="") tree.preorder(tree.root)

print("\nInorder Traversal: ", end="") tree.inorder(tree.root)

print("\nPostorder Traversal: ", end="") tree.postorder(tree.root)

print("\nLevel-order Traversal: ", end="") tree.level_order()

print()
```

OUTPUT:

```
========== RESTART: /Users/vamsi/Documents/DSA CSE25154/Lab8(b)btll.py ==========
Binary Tree (Preorder): A B D E C F
Preorder Traversal: A B D E C F
Inorder Traversal: D B E A C F
Postorder Traversal: D E B F C A
Level-order Traversal: A B C D E F
>>
```

EXPLANATION:

- The Node class creates a node containing data, left, and right links.
- The BinaryTree class manages the binary tree using linked nodes.
- The __init__() function initializes the tree with root = None.
- The set_root() function creates the root node.
- The set_left() function inserts the left child of a parent node.
- The set_right() function inserts the right child of a parent node.
- The search() function searches for a node using a queue.
- The preorder() function traverses in the order Root ---> Left---> Right.
- The inorder() function traverses in the order Left ---> Root ---> Right.
- The postorder() function traverses in the order Left ---> Right ---> Root.
- The level_order() function visits nodes level by level using a queue.
- The display() function displays the tree using preorder traversal.
- The main program creates a binary tree with nodes A, B, C, D, E, F and performs all four traversals.
- The program demonstrates binary tree creation, insertion, searching, and traversal using a linked list.

ERROR TABLE:

S.no.	Errors	Rectification
1.	**AttributeError** occurred when accessing node links incorrectly.	Used node.left and node.right correctly.
2.	Root already exists error occurred when trying to insert another root.	Checked whether self.root is None before insertion.
3.	Child already exists error occurred when inserting a child into an occupied position.	Checked whether node.left or node.right is None before insertion.
4.	Incorrect traversal output occurred when traversal order was incorrect.	Followed the correct preorder, inorder, postorder, and level-order sequences.

Week-9

AIM: Write a python program to implement a binary search tree with its operations traversals

PROGRAM:

```
class Node:

    def__init__(self, data):

        self.data = data

        self.left = None

        self.right = None
```

Insert a node

```
def insert(root, x): if root is None: return Node(x)

if x < root.data:
   root.left = insert(root.left, x)
elif x > root.data:
   root.right = insert(root.right, x)
else:
   print("Duplicate value is not allowed.")


return root
```

Inorder Traversal: Left, Root, Right

```
def inorder(root): if root is not None: inorder(root.left) print(root.data, end=" ")
inorder(root.right)
```

Preorder Traversal: Root, Left, Right

```
def preorder(root): if root is not None: print(root.data, end=" ") preorder(root.left)
preorder(root.right)
```

Postorder Traversal: Left, Right, Root

```
def postorder(root): if root is not None: postorder(root.left) postorder(root.right)
print(root.data, end=" ")
```

Find minimum element

```
def find_min(root): if root is None: return None

while root.left is not None:
    root = root.left

return root
```

Find maximum element

```
def find_max(root): if root is None: return None

while root.right is not None:
    root = root.right

return root
```

Search for an element

```
def search(root, x): if root is None: return None

if x < root.data:
    return search(root.left, x)
elif x > root.data:
    return search(root.right, x)
else:
    return root
```

Count total nodes

```
def count_nodes(root): if root is None: return 0

return 1 + count_nodes(root.left) + count_nodes(root.right)
```

Count leaf nodes

```
def count_leaves(root): if root is None: return 0

if root.left is None and root.right is None:
    return 1

return count_leaves(root.left) + count_leaves(root.right)
```

Count internal nodes

```
def count_internal(root): if root is None: return 0

if root.left is None and root.right is None:
    return 0
```

```
return 1 + count_internal(root.left) + count_internal(root.right)
```

Find height of the tree

Height is measured in edges; an empty tree has height -1.

```
def height(root): if root is None: return -1

return 1 + max(height(root.left), height(root.right))
```

Delete a node

```
def delete (root, x): if root is None: return None

if x < root.data:
    root.left = delete(root.left, x)

elif x > root.data:
    root.right = delete(root.right, x)

else:
    # Case 1: No children
    if root.left is None and root.right is None:
        return None

    # Case 2: Only right child
    elif root.left is None:
        return root.right

    # Case 3: Only left child
    elif root.right is None:
        return root.left

    # Case 4: Two children
    successor = find_min(root.right)
    root.data = successor.data
    root.right = delete(root.right, successor.data)

return root
```

Level-order Traversal using a queue

```
def level_order(root): if root is None: return

queue = [root]
front = 0

while front < len(queue):
```

```
    node = queue[front]
    front += 1

    print (node.data, end=" ")

    if node.left is not None:
        queue.append(node.left)

    if node.right is not None:
        queue.append(node.right)
```

Main menu-driven program

```
root = None

while True:

    print ("\n\n===== BINARY SEARCH TREE =====")

    print ("1. Insert")

    print ("2. Preorder Traversal")

    print ("3. Inorder Traversal")

    print ("4. Postorder Traversal")

    print ("5. Find Minimum")

    print ("6. Find Maximum")

    print ("7. Search")

    print ("8. Delete")

    print ("9. Count Total Nodes")

    print ("10. Count Leaf Nodes")

    print ("11. Count Internal Nodes")

    print ("12. Find Height")

    print ("13. Level Order Traversal")

    print ("14. Exit")
```

```
 try:
    choice = int (input ("Enter your choice: ")) except ValueError:
    print ("Please enter a valid integer choice.")
    continue

if choice == 1:
    try:
        data = int(input("Enter the data to insert: "))
        existed = search(root, data) is not None
        root = insert(root, data)

        if not existed:
            print("Element inserted successfully.")
    except ValueError:
        print("Please enter an integer.")

elif choice == 2:
    print("Preorder Traversal:", end=" ")
    if root is None:
        print("Tree is empty.")
    else:
        preorder(root)
        print()

elif choice == 3:
    print("Inorder Traversal:", end=" ")
    if root is None:
        print("Tree is empty.")
    else:
        inorder(root)
        print()

elif choice == 4:
    print("Postorder Traversal:", end=" ")
    if root is None:
        print("Tree is empty.")
    else:
        postorder(root)
        print()

elif choice == 5:
    node = find_min(root)
    if node is None:
```

```
            print("Tree is empty.")
        else:
            print("Minimum element:", node.data)

    elif choice == 6:
        node = find_max(root)
        if node is None:
            print("Tree is empty.")
        else:
            print("Maximum element:", node.data)

    elif choice == 7:
        try:
            data = int(input("Enter the element to search: "))
            if search(root, data) is not None:
                print("Element", data, "is found.")
            else:
                print("Element", data, "is not found.")
        except ValueError:
            print("Please enter an integer.")

    elif choice == 8:
        try:
            data = int(input("Enter the element to delete: "))
            if search(root, data) is None:
                print("Element not found.")
            else:
                root = delete(root, data)
                print("Element deleted successfully.")
        except ValueError:
            print("Please enter an integer.")

    elif choice == 9:
        print("Number of nodes:", count_nodes(root))

    elif choice == 10:
        print("Number of leaf nodes:", count_leaves(root))

    elif choice == 11:
        print("Number of internal nodes:", count_internal(root))

    elif choice == 12:
        print("Height of BST:", height(root))
```

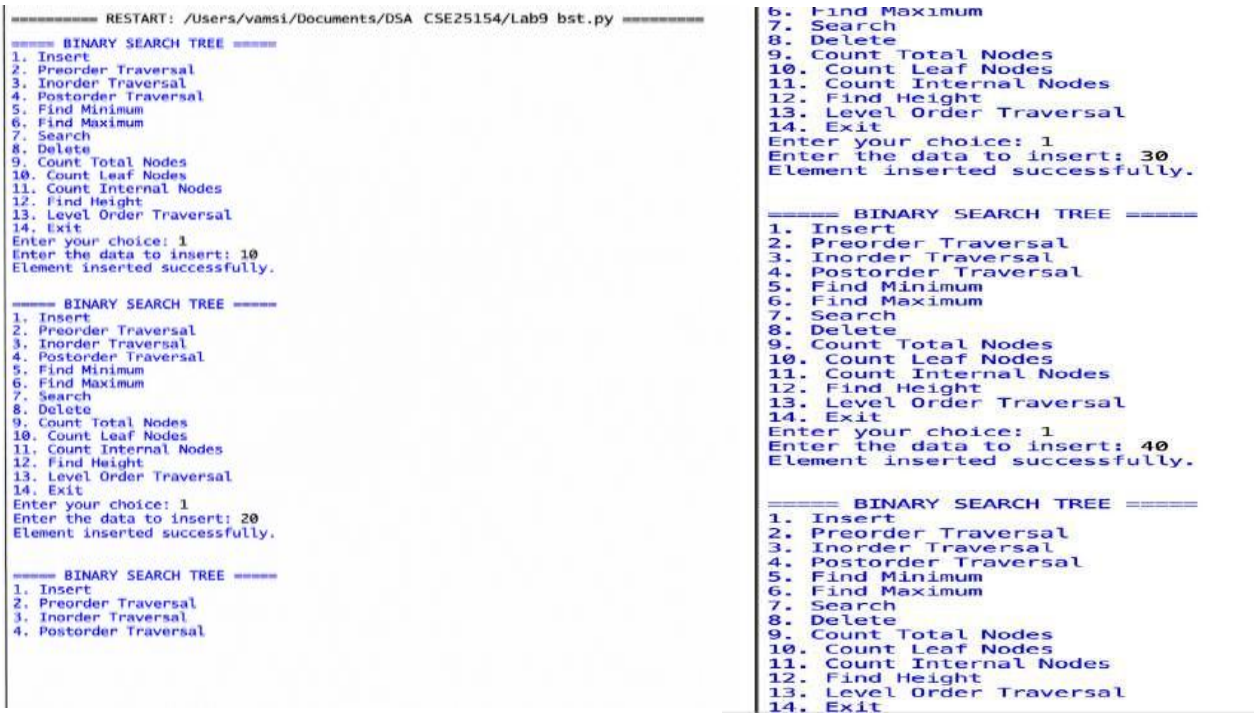
```
elif choice == 13:
    print("Level Order Traversal:", end=" ")
    if root is None:
        print("Tree is empty.")
    else:
        level_order(root)
        print()

elif choice == 14:
    print("Program terminated.")
    break

else:

print("Invalid choice. Please try again.")
```

OUTPUT:



```
Enter the data to insert: 50
Element inserted successfully.

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 2
Preorder Traversal: 10 20 30 40 50

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 3
Inorder Traversal: 10 20 30 40 50

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
```

```
13. Level Order Traversal
14. Exit
Enter your choice: 4
Postorder Traversal: 50 40 30 20 10

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 5
Minimum element: 10

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 6
Maximum element: 50

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
```

```
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 7
Enter the element to search: 40
Element 40 is found.

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 7
Enter the element to search: 100
Element 100 is not found.

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
```

```
Enter the element to delete: 50
Element deleted successfully.

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 99
Invalid choice. Please try again.

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 9\
Please enter a valid integer choice.

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
```

```
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 9
Number of nodes: 4

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 10
Number of leaf nodes: 1

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 11
Number of internal nodes: 3
```

```
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 12
Height of BST: 3

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 13
Level Order Traversal: 10 20 30 40

===== BINARY SEARCH TREE =====
1. Insert
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Find Minimum
6. Find Maximum
7. Search
8. Delete
9. Count Total Nodes
10. Count Leaf Nodes
11. Count Internal Nodes
12. Find Height
13. Level Order Traversal
14. Exit
Enter your choice: 14
Program terminated.
```

EXPLANATION:

- The Node class creates a node containing data, a left link, and a right link.
- The insert() function inserts values into the BST, placing smaller values on the left and larger values on the right.
- The inorder() function traverses the tree in the order Left → Root → Right.
- The preorder() function traverses the tree in the order Root → Left → Right.
- The postorder() function traverses the tree in the order Left → Right → Root.
- The find_min() function finds the minimum element by moving to the leftmost node.
- The find_max() function finds the maximum element by moving to the rightmost node.The search() function searches for an element using BST comparisons.
- The count_nodes() function counts the total number of nodes. The count_leaves() function counts nodes that have no children.The count_internal() function counts nodes that have at least one child.
- The height() function calculates the height of the tree in edges.
- The delete() function removes a node, handling cases with no children, one child, or two children.The level_order() function visits nodes level by level using a queue.
- The main menu allows the user to select and perform each BST operation, validate inputs, and exit the program.
- The program demonstrates BST insertion, deletion, searching, node counting, height calculation, and four traversal methods.

ERROR TABLE:

S.no.	Errors	Rectification
1.	TypeError / ValueError occurred when invalid input was entered.	Used try-except to validate integer inputs.
2.	Duplicate value error occurred when inserting an existing value.	Checked for duplicate values before insertion.
3.	Search error occurred when the element was not present in the BST.	Checked whether search() returns None before displaying the result.
4.	Empty tree error occurred when finding the minimum or maximum of an empty tree.	Checked whether the root is None before performing the operation.
5.	Deletion error occurred when deleting a node that did not exist.	Searched for the element before calling the delete() function.
6.	Incorrect traversal output occurred when the traversal order was incorrect.	Followed the correct preorder, inorder, postorder, and level-order sequences

Week-10

AIM: Write a python program to implement a AVL tree with all its operations.

PROGRAM:

```
class Node:

    def __init__(self, key):

        self.key = key

        self.left = None

        self.right = None

        self.height = 1

class AVLTree:

# Get height of a node

def height(self, node):

    if node is None:

        return 0

    return node.height

# Calculate balance factor
def get_balance(self, node):
    if node is None:
        return 0
    return self.height(node.left) - self.height(node.right)
# Right rotation
def right_rotate(self, y):
    x = y.left
    T2 = x.right
    x.right = y
    y.left = T2

    y.height = 1 + max(
        self.height(y.left), self.height(y.right)
    )
```

```
    x.height = 1 + max(
        self.height(x.left), self.height(x.right)
    )

    return x

# Left rotation
def left_rotate(self, x):
    y = x.right
    T2 = y.left

    y.left = x
    x.right = T2

    x.height = 1 + max(
        self.height(x.left), self.height(x.right)
    )
    y.height = 1 + max(
        self.height(y.left), self.height(y.right)
    )

    return y

# Insert a key
def insert(self, root, key):
    if root is None:
        return Node(key)

    if key < root.key:
        root.left = self.insert(root.left, key)
    elif key > root.key:
        root.right = self.insert(root.right, key)
    else:
        return root  # Ignore duplicate keys

    root.height = 1 + max(
        self.height(root.left), self.height(root.right)
    )

    balance = self.get_balance(root)

    # LL Case
    if balance > 1 and key < root.left.key:
```

```
            return self.right_rotate(root)

        # RR Case
        if balance < -1 and key > root.right.key:
            return self.left_rotate(root)

        # LR Case
        if balance > 1 and key > root.left.key:
            root.left = self.left_rotate(root.left)
            return self.right_rotate(root)

        # RL Case
        if balance < -1 and key < root.right.key:
            root.right = self.right_rotate(root.right)
            return self.left_rotate(root)

        return root

    # Find node with minimum key
    def min_value_node(self, node):
        current = node
        while current.left is not None:
            current = current.left
        return current

    # Find node with maximum key
    def max_value_node(self, node):
        current = node
        while current.right is not None:
            current = current.right
        return current

    # Delete a key
    def delete(self, root, key):
        if root is None:
            return root

        if key < root.key:
            root.left = self.delete(root.left, key)
        elif key > root.key:
            root.right = self.delete(root.right, key)
        else:
            # Node with zero or one child
```

```
            if root.left is None:
                return root.right
            elif root.right is None:
                return root.left

            # Node with two children:
            # Replace with inorder successor
            successor = self.min_value_node(root.right)
            root.key = successor.key
            root.right = self.delete(root.right, successor.key)

        # Update height
        root.height = 1 + max(
            self.height(root.left), self.height(root.right)
        )

        balance = self.get_balance(root)

        # LL Case
        if balance > 1 and self.get_balance(root.left) >= 0:
            return self.right_rotate(root)

        # LR Case
        if balance > 1 and self.get_balance(root.left) < 0:
            root.left = self.left_rotate(root.left)
            return self.right_rotate(root)

        # RR Case
        if balance < -1 and self.get_balance(root.right) <= 0:
            return self.left_rotate(root)

        # RL Case
        if balance < -1 and self.get_balance(root.right) > 0:
            root.right = self.right_rotate(root.right)
            return self.left_rotate(root)

        return root

    # Search for a key
    def search(self, root, key):
        if root is None or root.key == key:
            return root
```

```
    if key < root.key:
        return self.search(root.left, key)

    return self.search(root.right, key)

# Inorder traversal: Left, Root, Right
def inorder(self, root):
    if root:
        self.inorder(root.left)
        print(root.key, end=" ")
        self.inorder(root.right)

# Preorder traversal: Root, Left, Right
def preorder(self, root):
    if root:
        print(root.key, end=" ")
        self.preorder(root.left)
        self.preorder(root.right)

# Postorder traversal: Left, Right, Root
def postorder(self, root):
    if root:
        self.postorder(root.left)
        self.postorder(root.right)
        print(root.key, end=" ")

# Level-order traversal
def level_order(self, root):
    if root is None:
        print("Tree is empty")
        return

    queue = [root]

    while queue:
        node = queue.pop(0)
        print(node.key, end=" ")

        if node.left:
            queue.append(node.left)
        if node.right:
            queue.append(node.right)
```

```
# Display tree sideways
def display(self, root, level=0):
    if root:
        self.display(root.right, level + 1)
        print("     " * level + str(root.key))
        self.display(root.left, level + 1)
```

---------- MAIN PROGRAM ----------

```
tree = AVLTree()

root = None

while True:

        print("\n\n--- AVL TREE OPERATIONS ---")

        print("1. Insert")

        print("2. Delete")

        print("3. Search")

        print("4. Inorder Traversal")

        print("5. Preorder Traversal")

        print("6. Postorder Traversal")

        print("7. Level Order Traversal")

        print("8. Find Minimum")

        print("9. Find Maximum")

        print("10. Find Height")

        print("11. Display Tree")

        print("12. Exit")

choice = input("Enter your choice: ").strip()

if choice == "1":
    try:
        key = int(input("Enter value to insert: "))
```

```
            root = tree.insert(root, key)

            print("Insertion completed.")
        except ValueError:
            print("Please enter a valid integer.")

    elif choice == "2":
        try:
            key = int(input("Enter value to delete: "))
            if tree.search(root, key):
                root = tree.delete(root, key)
                print("Deletion completed.")
            else:
                print("Value not found.")
        except ValueError:
            print("Please enter a valid integer.")

    elif choice == "3":
        try:
            key = int(input("Enter value to search: "))
            if tree.search(root, key):
                print("Value found.")
            else:
                print("Value not found.")
        except ValueError:
            print("Please enter a valid integer.")

    elif choice == "4":
        print("Inorder:", end=" ")
        tree.inorder(root)
        print()

    elif choice == "5":
        print("Preorder:", end=" ")
        tree.preorder(root)
        print()

    elif choice == "6":
        print("Postorder:", end=" ")
        tree.postorder(root)
        print()

    elif choice == "7":
```

```
    print("Level order:", end=" ")
    tree.level_order(root)
    print()

elif choice == "8":
    if root:
        print("Minimum value:", tree.min_value_node(root).key)
    else:
        print("Tree is empty.")

elif choice == "9":
    if root:
        print("Maximum value:", tree.max_value_node(root).key)
    else:
        print("Tree is empty.")

elif choice == "10":
    print("Tree height:", tree.height(root))

elif choice == "11":
    if root:
        print("AVL Tree:")
        tree.display(root)
    else:
        print("Tree is empty.")

elif choice == "12":
    print("Exiting program.")
    break

else:
    print("Invalid choice. Try again.")
```

OUTPUT:

```
Python 3.13.5 (v3.13.5:6cb20a219a8, Jun 11 2025, 12:23:45) [Clang 16.0.0 (clang-
Enter "help" below or click "Help" above for more information.

======= RESTART: /Users/vamsi/Documents/DSA          CSE25154/Lab10      Avl trees.py =======


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 1
Enter value to insert: 30
Insertion completed.


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 1
Enter value to insert: 20
Insertion completed.


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
```

```
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 6
Postorder: 10 30 20


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 7
Level order: 20 10 30


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 8
Minimum value: 10


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
```

```
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 1
Enter value to insert: 10
Insertion completed.


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 4
Inorder: 10 20 30


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 5
Preorder: 20 10 30


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
```

```
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 9
Maximum value: 30


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 10
Tree height: 2


--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 11
AVL Tree:
    30
20
    10
```

```
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 3
Enter value to search: 30
Value found.

--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 3
Enter value to search: 10
Value not found.

--- AVL TREE OPERATIONS ---
1. Insert
2. Delete
3. Search
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Level Order Traversal
8. Find Minimum
9. Find Maximum
10. Find Height
11. Display Tree
12. Exit
Enter your choice: 12
Exiting program.
```

EXPLANATION:

- The Node class creates a node containing a key, left link, right link, and height.
- The height() function returns the height of a node and returns 0 if the node is empty.
- The get_balance() function calculates the balance factor using the heights of the left and right subtrees.
- The right_rotate() and left_rotate() functions perform rotations to balance the AVL tree.
- The insert() function inserts values into the AVL tree, placing smaller values on the left and larger values on the right.
- The LL, RR, LR, and RL cases handle tree imbalance using appropriate rotations.
- The min_value_node() function finds the minimum element by moving to the leftmost node.
- The max_value_node() function finds the maximum element by moving to the rightmost node.
- The delete() function removes a node, handling cases with no children, one child, or two children, and balances the tree.
- The search() function searches for an element using BST comparisons.
- The inorder() function traverses the tree in the order Left → Root → Right.
- The preorder() function traverses the tree in the order Root → Left → Right.
- The postorder() function traverses the tree in the order Left → Right → Root.
- The level_order() function visits nodes level by level using a queue.
- The display() function displays the AVL tree structure sideways.

- The main menu allows the user to select and perform each AVL tree operation, validate inputs, and exit the program.
- The program demonstrates AVL tree insertion, deletion, searching, balancing, minimum and maximum element finding, height calculation, and traversal methods.

ERROR TABLE:

S.no.	Errors	Rectification
1.	ValueError occurred when non-integer input was entered.	Used try-except to validate integer inputs.
2.	Duplicate values were entered during insertion.	Checked for duplicate keys before inserting a new node.
3.	Search failed when the given element was not present in the tree.	Checked whether search() returns None before displaying the result.
4.	Minimum or maximum value could not be found when the tree was empty.	Checked whether the root is None before performing the operation.
5.	Deletion failed when the given element was not present in the tree.	Searched for the element before calling the delete() function.
6.	Tree became unbalanced after insertion or deletion.	Applied appropriate LL, RR, LR and RL rotations.
7.	Traversal operations produced no elements when the tree was empty.	Checked whether the root is None before performing traversal.
8.	Incorrect height was obtained after tree modifications.	Updated node heights after insertion, deletion and rotations.

Week-11

AIM: Write a python program to implement Hashing (Seperate chaining, Linear probing, and Quadratic probing).

PROGRAM:

```
size = 10

#Separate Chaining

def separate_chaining(keys): table = [[] for i in range(size)]

for key in keys:
    index = key % size
    table[index].append(key)

print("\n separate Chaining:")
for i in range(size):
    print (i, ":", table[i])


#Linear Probing

def linear_probing(keys): table = [None] * size

for key in keys:
    index = key % size

    while table[index] is not None:
        index = (index + 1) % size

    table[index] = key

print("\nLinear Probing:")
for i in range(size):
    print(i, ":", table[i])


#Quadratic Probing

def quadratic_probing(keys): table = [None] * size

for key in keys:
    index = key % size
```

```
    i = 0

    while i < size:
        new_index = (index + i * i) % size

        if table[new_index] is None:
            table[new_index] = key
            break

        i += 1

print("\n Quadratic Probing:")
for i in range(size):
    print (i, ":", table[i])
 #Main program

keys = list (map (int, input ("Enter the keys: "). split ()))

separate_chaining(keys) linear_probing(keys) quadratic_probing(keys)
```

OUTPUT:

```
Python 3.13.5 (v3.13.5:6cb20a219a8, Jun 11 2025, 12:23:45) [Clang 16.0.0 (clang-1600.0.26.6)] on darwin
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: /Users/vamsi/Documents/DSA          CSE25154/Lab11      hashing.py ========
Enter the keys: 12 22 32 15 25 35 45

Separate Chaining:
0 : []
1 : []
2 : [12, 22, 32]
3 : []
4 : []
5 : [15, 25, 35, 45]
6 : []
7 : []
8 : []
9 : []

Linear Probing:
0 : None
1 : None
2 : 12
3 : 22
4 : 32
5 : 15
6 : 25
7 : 35
8 : 45
9 : None

Quadratic Probing:
0 : None
1 : 45
2 : 12
3 : 22
4 : 35
5 : 15
6 : 32
7 : None
8 : None
9 : 25
>>>
```

EXPLANATION:

- The size variable defines the size of the hash table as 10.
- The separate_chaining() function creates a hash table with empty lists to store the elements.
- The hash function key % size calculates the index for each key. The append() function inserts keys into the corresponding list. If multiple keys have the same index, they are stored in the same list.
- The linear_probing() function creates an empty hash table using None to represent empty positions.
- The while loop checks whether the calculated index is occupied. If a collision occurs, (index + 1) % size searches for the next available position and inserts the key.
- The quadratic_probing() function resolves collisions by checking alternative positions using square increments.
- The expression (index + i * i) % size calculates the new index. The if condition checks whether the position is empty, and the break statement stops the loop after insertion.
- The i += 1 statement increases the counter to check the next position if a collision occurs.
- The main program accepts multiple integer keys from the user using input (), split (), map (), and list ().
- The three functions are called to perform Separate Chaining, Linear Probing, and Quadratic Probing.
- The program demonstrates hashing, collision detection, and collision resolution using three different techniques.

ERROR TABLE:

S.no.	Errors	Rectification
1.	Collision occurred when multiple keys generated the same index.	Applied separate chaining and probing techniques.
2.	Index error occurred when the table reached its last position.	Used modulo operation to wrap around the table.
3.	Insertion failed when the hash table was full.	Checked for available positions before insertion.
4.	Elements were stored at incorrect positions during linear probing.	Used (index + 1) % size to find the next position.
5.	Elements were stored at incorrect positions during linear probing.	Used (index + 1) % size to find the next position.
6.	Incorrect positions were checked during quadratic probing.	Used the formula (index + i * i) % size.